



# Aakash

Medical | IIT-JEE | Foundations

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## FINAL TEST SERIES for NEET-2024

MM : 720

### Test -3 (Home Assignment)

Time : 3 Hrs. 20 Mins.

#### Topics covered :

- Physics** : Gravitation, Mechanical Properties of Solids, Mechanical Properties of Fluids, Thermal Properties of Matter  
**Chemistry** : Equilibrium, Redox Reactions, The *p*-Block Elements (Group-13 & 14)  
**Botany** : Cell: The Unit of Life, Cell Cycle and Cell Division  
**Zoology** : Breathing and Exchange of Gases, Body Fluids and Circulation

#### Instructions :

- There are two sections in each subject, i.e. Section-A & Section-B. You have to attempt all 35 questions from Section-A & only 10 questions from Section-B out of 15.
- Each question carries 4 marks. For every wrong response 1 mark shall be deducted from the total score. Unanswered / unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white-fluid or any other rubbing material on the Answer sheet.

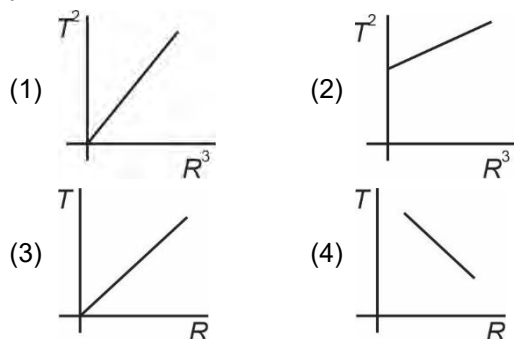
### PHYSICS

Choose the correct answer :

#### SECTION - A

- Unit of the quantity  $g/G$  in S.I system will be (Here symbols have their usual meaning)
  - $\text{kg/m}^2$
  - $\text{m/kg}^2$
  - $\text{m}^2/\text{kg}$
  - $\text{kg}^2/\text{m}$
- Three masses  $m$ ,  $2m$  and  $3m$  are placed at the three vertices of an equilateral triangle of side  $a$ . The gravitational potential energy of the system will be
  - $\frac{-9Gm^2}{a}$
  - $\frac{-11Gm^2}{a}$
  - $\frac{-7Gm^2}{a}$
  - $\frac{-5Gm^2}{a}$
- The kinetic energy of a satellite in its orbit around the earth is  $E$ . The additional kinetic energy given to the satellite so as to enable to escape from gravitational pull of the earth is
  - $4E$
  - $2E$
  - $\sqrt{2}E$
  - $E$

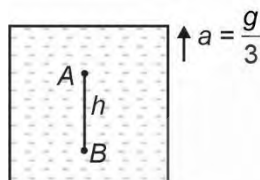
4. Which of the following graph is correct between time period of revolution ( $T$ ) and its radius of orbit ( $R$ ) for the motion of a satellite revolving about a planet?



5. 30% volume of a floating cubical block is outside the water. The relative density of the block is

- (1) 0.3 (2) 0.6  
(3) 0.4 (4) 0.7

6. A container filled with liquid is moving upwards with acceleration  $\frac{g}{3}$  and distance  $A$  and  $B$  is  $h$ . If density of liquid in the container is  $\rho$ , then pressure difference between  $A$  and  $B$  is



- (1)  $\frac{\rho gh}{3}$  (2)  $\frac{2\rho gh}{3}$   
(3)  $\frac{4\rho gh}{3}$  (4)  $\rho gh$

7. Water droplets are spherical in free fall condition due to

- (1) Gravity  
(2) Viscosity  
(3) Air drag  
(4) Surface tension

8. If two wires of same material and of same area of cross-section have length ratio 2 : 3, then under the effect of same stretching force their respective elongation will be in ratio

- (1) 1 : 1 (2) 4 : 9  
(3) 2 : 3 (4) 3 : 4

9. Longitudinal strain is meaningful in

- (1) Liquid (2) Gases  
(3) Solid (4) Both (1) and (3)

10. **Statement (A):** Steel is more elastic than rubber.

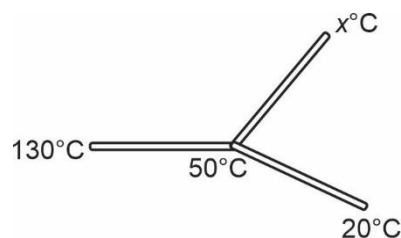
**Statement (B):** Upto proportional limit Hooke's law is applicable.

- (1) Both statement (A) and statement (B) are correct  
(2) Both statement (A) and statement (B) are incorrect  
(3) Statement (A) is correct and statement (B) is incorrect  
(4) Statement (A) is incorrect and statement (B) is correct

11. The 50 parts of Kelvin scale is equal to 'n' parts of Fahrenheit scale, where 'n' is

- (1) 50 (2) 90  
(3) 100 (4) 180

12. Three identical rods made of same material are joined together as shown in figure. The value of  $x$  is



- (1) 30 (2) 10  
(3) 20 (4) Zero

Space for Rough Work

13. Surface tension of water is  $0.072 \text{ N/m}$ . The excess pressure inside the water drop of radius  $1.2 \text{ mm}$  is  
 (1)  $60 \text{ N/m}^2$  (2)  $240 \text{ N/m}^2$   
 (3)  $180 \text{ N/m}^2$  (4)  $120 \text{ N/m}^2$
14. Two capillary tubes of radii  $0.2 \text{ cm}$  and  $0.4 \text{ cm}$  are dipped in same liquid. The ratio of height through which liquid will rise in the tubes is  
 (1)  $2 : 3$  (2)  $1 : 4$   
 (3)  $4 : 1$  (4)  $2 : 1$
15. The working of venturimeter is based on  
 (1) Torricelli's law (2) Bernoulli's theorem  
 (3) Archimedes principle (4) Stoke's law
16. A cylindrical vessel is filled with water upto height  $H$ . A hole is bored in the wall at depth  $h$  from the free surface of water. For maximum range  $h$  should be equal to  
 (1)  $\frac{H}{4}$  (2)  $\frac{3H}{4}$   
 (3)  $\frac{H}{2}$  (4)  $H$
17. The escape velocity of an object from the surface of a planet is  $v_e$ . A particle starts from rest at a large distance from the planet, reaches the planet only under gravitational attraction, and passes through a smooth tunnel through its centre. Its speed at the centre of the planet will be  
 (1)  $\sqrt{1.5}v_e$  (2)  $\frac{v_e}{\sqrt{2}}$   
 (3)  $v_e$  (4) Zero
18. Three identical point masses, each of mass  $1 \text{ kg}$  lie in the  $x$ - $y$  plane at points  $(0, 0)$   $(0, 2 \text{ m})$  and  $(2 \text{ m}, 0)$ . The gravitational force on the mass at the origin is  
 (1)  $\frac{G}{4}(-\hat{i} - \hat{j})$  (2)  $\frac{G}{4}(-\hat{i} + \hat{j})$   
 (3)  $\frac{G}{4}(\hat{i} + \hat{j})$  (4)  $\frac{G}{4}(\hat{i} - \hat{j})$
19. A tank full of water has a small hole at its bottom. Let  $t_1$  be the time taken to empty first one third of the tank and  $t_2$  be the time taken to empty second one third of the tank and  $t_3$  be the time taken to empty rest of the tank then  
 (1)  $t_1 = t_2 = t_3$   
 (2)  $t_1 > t_2 > t_3$   
 (3)  $t_1 < t_2 < t_3$   
 (4)  $t_1 > t_2 < t_3$
20. A piece of cork when released from rest at the bottom of a deep lake, it moves up. Its velocity  $v$  is plotted against time  $t$ . Which of the following best represents the resulting curve?
- 
21. A tank of height  $5 \text{ m}$  is full of water. There is a hole of cross sectional area  $1 \text{ cm}^2$  in its bottom. The initial volume of water that will come out from this hole per second is ( $g = 10 \text{ m/s}^2$ )  
 (1)  $10^{-3} \text{ m}^3/\text{s}$  (2)  $10^{-4} \text{ m}^3/\text{s}$   
 (3)  $10 \text{ m}^3/\text{s}$  (4)  $10^{-2} \text{ m}^3/\text{s}$
22. A soap bubble in vacuum has a radius of  $3 \text{ cm}$  and another soap bubble in vacuum has a radius of  $4 \text{ cm}$ . If the two bubbles coalesce under isothermal condition, then the radius of the new bubble is  
 (1)  $2.3 \text{ cm}$  (2)  $4.5 \text{ cm}$   
 (3)  $5 \text{ cm}$  (4)  $7 \text{ cm}$

Space for Rough Work

23. A constant volume gas thermometer shows pressure reading of 50 cm and 90 cm of mercury at  $0^\circ\text{C}$  and  $100^\circ\text{C}$  respectively. When the pressure reading is 60 cm of mercury, the temperature is

(1)  $25^\circ\text{C}$   
 (2)  $40^\circ\text{C}$   
 (3)  $15^\circ\text{C}$   
 (4)  $12.5^\circ\text{C}$

24. The coefficient of volume expansion of a liquid is  $49 \times 10^{-5} \text{ K}^{-1}$ . The fractional change in its density when the temperature is raised by  $30^\circ\text{C}$ , will be nearly

(1)  $7.5 \times 10^{-2}$  (2)  $3.0 \times 10^{-2}$   
 (3)  $1.5 \times 10^{-2}$  (4)  $1.1 \times 10^{-2}$

25. Surface of the lake is at  $2^\circ\text{C}$ . Find the temperature of the bottom of the lake

(1)  $2^\circ\text{C}$  (2)  $3^\circ\text{C}$   
 (3)  $4^\circ\text{C}$  (4)  $1^\circ\text{C}$

26. Consider a compound slab consisting of two different materials having equal thickness and thermal conductivities  $k$  and  $3k$  respectively as shown in figure. The equivalent thermal conductivity of the slab is

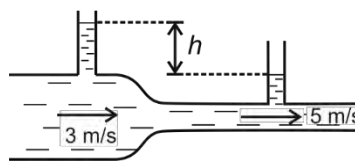


(1)  $\frac{3k}{2}$  (2)  $\frac{4k}{3}$   
 (3)  $\frac{k}{2}$  (4)  $4k$

27. The following four wires are made of the same material. Which of these will have the largest extension when the same tension is applied?

(1) Length 50 cm and diameter 2 mm  
 (2) Length 100 cm and diameter 1 mm  
 (3) Length 200 cm and diameter 2 mm  
 (4) Length 300 cm and diameter 3 mm

28. An incompressible liquid is flowing through a horizontal tube in streamline manner as shown in figure. The value of  $h$  is (in meters)



(1) 1 (2) 0.8  
 (3) 3.8 (4) 4.2

29. The length of a metal wire is  $l_1$  when the tension in it is  $T_1$  and is  $l_2$  when the tension is  $T_2$ . The unstretched length of the wire is

(1)  $\frac{l_1 T_1 + l_2 T_2}{T_1 + T_2}$  (2)  $\frac{l_1 + l_2}{2}$   
 (3)  $\frac{l_1 T_2 - l_2 T_1}{T_2 - T_1}$  (4)  $\frac{l_1 T_2 - l_2 T_1}{T_2 + T_1}$

30. The Young's modulus of the material of a wire is numerically equal to the

(1) Stress required to increase its length four times  
 (2) Stress required to produce unit strain  
 (3) Strain produced in it  
 (4) Half the strain produced in it

31. At what height from the surface of earth, value of acceleration due to gravity becomes  $\frac{1}{9}$ th of its value at surface of earth? ( $R$  is radius of earth)

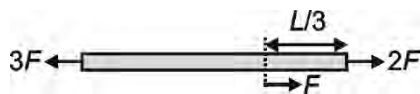
(1)  $3R$  (2)  $2R$   
 (3)  $R$  (4)  $\frac{R}{3}$

32. A lunar month of a planet consists of 84 days. If the distance between centres of planet and its moon becomes triple of its present value, then number of days in new lunar month will be close to

(1) 27 days (2) 848 days  
 (3) 172 days (4) 436 days

Space for Rough Work

33. A uniform cylindrical rod of length  $L$ , cross-sectional area  $A$  and Young's modulus  $Y$  is acted upon by the forces shown in the figure. The elongation of the rod is



- (1)  $\frac{3FL}{5AY}$  (2)  $\frac{2FL}{5AY}$   
 (3)  $\frac{3FL}{8AY}$  (4)  $\frac{8FL}{3AY}$
34. There is a horizontal film of soap solution. On it a thread is placed in the form of a loop. The film is pierced inside the loop and the thread becomes a circular loop of radius  $R$ . If the surface tension of the soap solution be  $T$ , then tension in the thread will be
- (1)  $\pi R^2 T$  (2)  $2RT$   
 (3)  $RT$  (4)  $\frac{\pi R^2}{T}$
35. A body is projected at an angle  $60^\circ$  from surface of Earth with velocity  $\sqrt{2gR}$ . Final kinetic energy of object is (symbols have usual meanings)
- (1) Zero (2)  $mgR$   
 (3)  $\frac{mgR}{2}$  (4) Infinite

### SECTION-B

36. In planetary motion, the angular momentum conservation leads to the law of
- (1) Orbit  
 (2) Areas  
 (3) Period  
 (4) Conservation of kinetic energy
37. The acceleration due to gravity is  $g$  at a point  $r$  distance from the centre of earth of radius  $R$ . If  $r < R$ , then

- (1)  $g \propto r$  (2)  $g \propto r^{-1}$   
 (3)  $g \propto r^{-2}$  (4)  $g \propto r^2$

38. If  $g$  is the acceleration due to gravity on the earth's surface, the gain in potential energy of an object of mass  $m$  raised from the surface of the earth to a height equal to radius  $R$  of the earth is

- (1)  $\frac{mgR}{2}$  (2)  $2mgR$   
 (3)  $mgR$  (4)  $\frac{mgR}{4}$

39. The escape velocity for a body projected vertically upward from the surface of earth is 11 km/s. If the body is projected at an angle of  $45^\circ$  with the vertical, then escape velocity will be

- (1) 11 km/s (2)  $11\sqrt{2}$  km/s  
 (3)  $\frac{11}{\sqrt{2}}$  km/s (4) 5 km/s

40. A cube is subjected to a uniform volume compression. If the side of the cube decreases by 1%, the bulk strain is

- (1) 0.03 (2) 0.06  
 (3) 0.02 (4) 0.01

41. The absorptance and reflectance of a material is 0.2 and 0.4 respectively. If heat incident on it is 400 J/s. Then energy transmitted in 10 s is

- (1) 160 J (2) 1600 J  
 (3) 2400 J (4) 240 J

42. Two liquids having masses in the ratio 2 : 1 at the temperature of  $30^\circ\text{C}$  and  $20^\circ\text{C}$  respectively are mixed with each other. If temperature of the mixture is  $26^\circ\text{C}$ , then ratio of their specific heat capacities  $S_1 : S_2$  is

- (1)  $\frac{4}{3}$  (2)  $\frac{3}{4}$   
 (3)  $\frac{3}{8}$  (4)  $\frac{8}{3}$

Space for Rough Work

43. The terminal velocity ( $v$ ) of a small sized spherical body of radius  $r$  falling vertically in a viscous liquid is such that

- (1)  $v \propto \frac{1}{r}$   
 (2)  $v \propto r$   
 (3)  $v \propto r^2$   
 (4)  $v \propto \frac{1}{r^2}$

44. When a capillary is dipped into a liquid, the levels of liquid inside and outside are equal. The angle of contact of the liquid inside the capillary is

- (1) Zero (2)  $90^\circ$   
 (3)  $45^\circ$  (4)  $60^\circ$

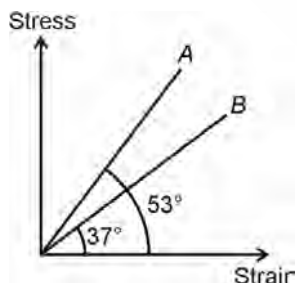
45. Consider a water droplet having radius  $10^{-2}$  m. It is broken into identical 1000 droplets. If surface tension of the water is 0.075 N/m then change in surface energy is

- (1)  $27\pi \times 10^{-5}$  J (2)  $30\pi \times 10^{-5}$  J  
 (3)  $27\pi \times 10^{-3}$  J (4)  $30\pi \times 10^{-3}$  J

46. If velocity head of a stream of water is equal to 10 cm, then its speed of flow is approximately

- (1) 1.0 m/s (2) 1.4 m/s  
 (3) 140 m/s (4) 10 m/s

47. The stress versus strain graphs for two wires made of different materials A and B are as shown in the figure. If  $Y_A$  and  $Y_B$  are the Young's moduli of the materials, then



(1)  $Y_B = 2Y_A$

(2)  $Y_A = Y_B$

(3)  $Y_B = 3Y_A$

(4)  $Y_B = \frac{9}{16}Y_A$

48. A bimetallic strip has two different metal bars 1 and 2 welded parallel to their length. Given thermal coefficient of linear expansion  $\alpha_1 > \alpha_2$  and length of bars is same. Upon cooling, the strip will bend with

- (1) Remains straight  
 (2) Concave on side 1  
 (3) Convex on side 1  
 (4) Any of the above is possible

49. A beaker is completely filled with water at  $4^\circ\text{C}$ . It will overflow if (ignore expansion of beaker)

- (1) Heated above  $4^\circ\text{C}$   
 (2) Cooled below  $4^\circ\text{C}$   
 (3) Both (1) and (2)  
 (4) It will not overflow at all

50. A planet of mass  $m$  revolves around the sun of mass  $M$  in a circular orbit of radius  $r$  with angular speed  $\omega$ . Another planet of mass  $2m$  revolves around the sun in circular orbit of radius  $4r$  with angular velocity  $\omega'$ . Then  $\frac{\omega}{\omega'}$  is

(1) 4

(2)  $\frac{1}{4}$

(3) 8

(4)  $\frac{1}{2}$

Space for Rough Work

# CHEMISTRY

## SECTION - A

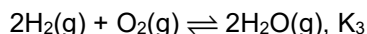
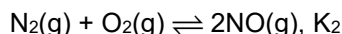
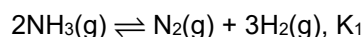
51. Which among the following aqueous salt solution is acidic in nature?

- (1) Sodium acetate
- (2) Potassium chloride
- (3) Ammonium sulphate
- (4) Sodium cyanide

52. Percentage ionization of 0.01 M acetic acid in 0.2 M HCl [ $K_a(\text{CH}_3\text{COOH}) = 1.8 \times 10^{-5}$ ] is

- (1) 0.09%                      (2) 0.009%
- (3) 0.9%                      (4) 9.1%

53. Consider the following equilibria with their equilibrium constants



The equilibrium constant (K) for the reaction  $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightleftharpoons 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{g})$  will be

- (1)  $K_1^2 K_2^2 K_3^3$                       (2)  $K_1^2 K_2^3 K_3^2$
- (3)  $K_1^3 K_2^2 K_3^2$                       (4)  $K_1^2 K_2^2 K_3^2$

54. For which of the following reversible reaction,  $K_P > K_C$ ?

- (1)  $2\text{NO}(\text{g}) \rightleftharpoons \text{N}_2(\text{g}) + \text{O}_2(\text{g})$
- (2)  $\text{NO}(\text{g}) + \text{O}_3(\text{g}) \rightleftharpoons \text{NO}_2(\text{g}) + \text{O}_2(\text{g})$
- (3)  $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
- (4)  $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$

55. The number of sigma and pi bonds present in borazine, respectively are

- (1) 12 and 3                      (2) 6 and 3
- (3) 9 and 4                      (4) 3 and 3

56. Given below are two statements:

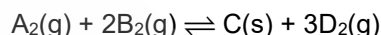
**Statement (I):** On increasing temperature, pOH of water decreases.

**Statement (II):** On increasing temperature,  $K_w$  of water decreases.

In the light of above statements, choose the **correct** option among the following.

- (1) Statement (I) is true but statement (II) is false
- (2) Statement (I) is false but statement (II) is true
- (3) Both statement (I) and statement (II) are true
- (4) Both statement (I) and statement (II) are false

57. Consider the reversible reaction



The correct expression for the equilibrium constant is

- (1)  $\frac{[\text{D}_2]^3}{[\text{A}_2] \times [\text{B}_2]^2}$                       (2)  $\frac{[\text{C}] \times [\text{D}_2]}{[\text{A}_2]^2 \times [\text{B}_2]}$
- (3)  $\frac{[\text{D}_2]^3}{2[\text{A}_2] \times [\text{B}_2]}$                       (4)  $\frac{[\text{A}_2][\text{B}_2]^2}{[\text{C}] \times 3[\text{D}_2]}$

58. The conjugate acid of  $\text{HSO}_4^-$  is

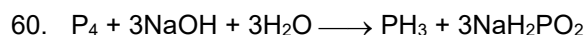
- (1)  $\text{SO}_4^{2-}$                       (2)  $\text{SO}_3$
- (3)  $\text{H}_2\text{SO}_4$                       (4)  $\text{H}_2\text{SO}_3$

59. 0.5 moles of  $\text{CH}_3\text{COONa}$  are mixed with 0.2 moles of HCl in an aqueous solution. The pH of the resulting solution will be

[Given  $\text{p}K_a$  of  $\text{CH}_3\text{COOH} = 4.74$ ;  $[\log 3 = 0.48]$

- (1) 5.12                      (2) 4.10
- (3) 4.92                      (4) 4.45

Space for Rough Work



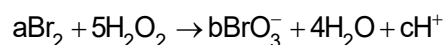
The equivalent mass of  $P_4$  in the above reaction is  
(M = Molecular weight of  $P_4$ )

- (1)  $\frac{M}{3}$  (2)  $\frac{M}{12}$   
(3)  $\frac{3M}{2}$  (4) M

61. Select the compound which can behave as a reducing agent only

- (1)  $SO_2$  (2)  $SO_3$   
(3)  $H_2S$  (4)  $HNO_2$

62. Consider the given reaction.



The values of a, b and c for the balanced redox reaction respectively are

- (1) 2, 1, 4 (2) 1, 2, 2  
(3) 3, 2, 1 (4) 1, 1, 2

63. Oxidation number of central carbon atom in  $C_3O_2$  is

- (1) Zero (2) +2  
(3) +1 (4) +4

64. Disproportionation reaction is not shown by

- (1)  $HClO$  (2)  $HClO_2$   
(3)  $HClO_3$  (4)  $HClO_4$

65. The number of mole(s) of  $KMnO_4$  that will be needed to react with one mole of  $Sn^{2+}$  ion in acidic solution is

- (1) 1 (2)  $\frac{2}{5}$   
(3)  $\frac{6}{5}$  (4) 2

66. Which of the following is a self-indicator?

- (1)  $MnO_4^-$  (2)  $S_2O_3^{2-}$   
(3)  $C_2O_4^{2-}$  (4)  $SO_4^{2-}$

67. Given the standard electrode potentials

$$K^+/K = -2.93 \text{ V}, Ag^+/Ag = 0.8 \text{ V},$$

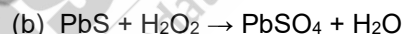
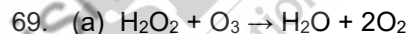
$$Zn^{2+}/Zn = -0.76 \text{ V and } Cr^{3+}/Cr = -0.74 \text{ V}$$

Select the **correct** statement.

- (1) Zn can reduce  $K^+$ ,  $Ag^+$  and  $Cr^{3+}$   
(2) Zn can reduce  $Ag^+$ ,  $Cr^{3+}$  but not  $K^+$   
(3) Zn can reduce  $K^+$  but not  $Ag^+$  and  $Cr^{3+}$   
(4) Zn can reduce  $K^+$ ,  $Cr^{3+}$  but not  $Ag^+$

68. In a closed system at equilibrium,  $NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$ , if the partial pressure of  $CO_2$  is doubled then partial pressure of  $NH_3(g)$  will become

- (1)  $3\sqrt{2}$  times of the original total pressure  
(2)  $\frac{1}{3\sqrt{2}}$  times of the original total pressure  
(3)  $\frac{\sqrt{2}}{3}$  times of the original total pressure  
(4)  $\frac{3}{\sqrt{2}}$  times of the original total pressure



Role of hydrogen peroxide in the above reactions is respectively

- (1) Oxidising in (a) and reducing in (b)  
(2) Reducing in (a) and oxidising in (b)  
(3) Reducing in both (a) and (b)  
(4) Oxidising in both (a) and (b)

70. In which among the following compounds, oxygen carry +2 oxidation state?

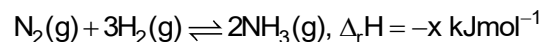
- (1)  $RbO_2$   
(2)  $O_2F_2$   
(3)  $OF_2$   
(4)  $H_2O_2$

Space for Rough Work

71. The solubility of CdS in 0.4 M Na<sub>2</sub>S solution would be ( $K_{sp}$  of CdS is  $8 \times 10^{-27}$ )

- (1)  $2 \times 10^{-26}$  mol/L (2)  $4 \times 10^{-14}$  mol/L  
(3)  $4 \times 10^{-26}$  mol/L (4)  $2 \times 10^{-27}$  mol/L

72. For the reaction,



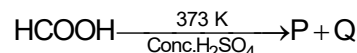
Which of the following statements is not true?

- (1) Addition of inert gas at constant volume will not have any effect on equilibrium  
(2) Addition of N<sub>2</sub>(g) at equilibrium will shift the reaction in backward direction  
(3) Decrease in temperature will shift the reaction to forward direction  
(4) The increase in pressure will shift the equilibrium to the right

73. Which of the following on hydrolysis followed by condensation polymerisation yields straight chain polymers?

- (1) MeSiCl<sub>3</sub> (2) Me<sub>2</sub>SiCl<sub>2</sub>  
(3) Me<sub>3</sub>SiCl (4) Me<sub>4</sub>Si

74. The product P and Q in the following reaction respectively are



- (1) H<sub>2</sub> and CO (2) H<sub>2</sub>O and CO<sub>2</sub>  
(3) H<sub>2</sub> and CO<sub>2</sub> (4) H<sub>2</sub>O and CO

75. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

**Assertion (A):** In H<sub>3</sub>PO<sub>4</sub>, higher order ionisation constants ( $K_{a2}$  and  $K_{a3}$ ) are smaller than lower order ionization constant ( $K_{a1}$ ).

**Reason (R):** In H<sub>3</sub>PO<sub>4</sub>, it is more difficult to remove a positively charged proton from a negative ions (H<sub>2</sub>PO<sub>4</sub><sup>-</sup> and HPO<sub>4</sub><sup>2-</sup>) due to electrostatic forces.

In the light of the above statements, select the **correct** option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)  
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

76. Aluminium chloride in acidified aqueous solutions forms a complex ion 'A' in which hybridisation state of Al is B. A and B respectively are

- (1) [Al(H<sub>2</sub>O)<sub>6</sub>]<sup>3+</sup>,  $sp^3d^2$   
(2) [Al(H<sub>2</sub>O)<sub>4</sub>]<sup>3+</sup>,  $dsp^3$   
(3) [Al(H<sub>2</sub>O)<sub>6</sub>]<sup>3+</sup>,  $d^2sp^3$   
(4) [Al(H<sub>2</sub>O)<sub>4</sub>]<sup>3+</sup>,  $sp^3$

77. Match **List-I** with **List-II** and select the correct option.

|    | List-I<br>Oxide                |       | List-II<br>Chemical nature |
|----|--------------------------------|-------|----------------------------|
| a. | CO                             | (i)   | Acidic                     |
| b. | In <sub>2</sub> O <sub>3</sub> | (ii)  | Amphoteric                 |
| c. | Ga <sub>2</sub> O <sub>3</sub> | (iii) | Basic                      |
| d. | SiO <sub>2</sub>               | (iv)  | Neutral                    |

(1) a(iv), b(iii), c(ii), d(i)

(2) a(i), b(ii), c(iii), d(iv)

(3) a(ii), b(iii), c(iv), d(i)

(4) a(iii), b(iv), c(i), d(ii)

78. Strongest Lewis acid among the following boron halides is

- (1) BF<sub>3</sub>  
(2) BCl<sub>3</sub>  
(3) BBr<sub>3</sub>  
(4) BI<sub>3</sub>

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79. Given below are two statements :

**Statement (I):** Graphite has layered structure.

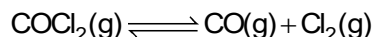
**Statement (II):** Carbon black is obtained by burning hydrocarbons in a limited supply of air.

In the light of above statements choose the most appropriate answer from the options given below.

- (1) Statement (I) is correct but statement (II) is incorrect  
 (2) Statement (I) is incorrect but statement (II) is correct  
 (3) Both Statements (I) and (II) are correct  
 (4) Both Statements (I) and (III) are incorrect
80. Oxidation number of chlorine in  $\text{CaOCl}_2$  is (are)  
 (1) -1 only (2) +1 only  
 (3) Both +1 and -1 (4) -2 only
81. Aqueous solution of which of the following act as buffer?  
 (1) NaCl (2)  $\text{NH}_4\text{Cl}$   
 (3)  $\text{NH}_4\text{CN}$  (4) KCl
82. The value of  $\Delta G^\circ$  at 300K if equilibrium constant for the reaction  $K_{\text{eq}} = 10^3$  is  
 (1)  $-4.14 \text{ kcal mol}^{-1}$  (2)  $-2.18 \text{ kcal mol}^{-1}$   
 (3)  $4.14 \text{ kcal mol}^{-1}$  (4)  $-41.4 \text{ cal mol}^{-1}$
83. Number of bridge B - H - B bonds in a  $\text{B}_2\text{H}_6$  molecule is  
 (1) 2 (2) 4  
 (3) 1 (4) 3
84. According to inert pair effect, Pb shows its more stable oxidation state as  
 (1) +1 (2) +2  
 (3) +3 (4) +4
85. Which among the following mixture of gases is called producer gas?  
 (1)  $\text{CO} + \text{H}_2$   
 (2)  $\text{CO} + \text{H}_2\text{O}$   
 (3)  $\text{CO} + \text{N}_2$   
 (4)  $\text{CO}_2 + \text{N}_2$

### SECTION-B

86. Following two equilibriums are simultaneously taking place in a vessel



If  $\text{CO}(\text{g})$  is introduced into the vessel then the amount of

- (1)  $\text{PCl}_5$  will remain same  
 (2)  $\text{PCl}_3$  will be decreased  
 (3)  $\text{PCl}_5$  will be decreased  
 (4)  $\text{COCl}_2$  will be decreased
87. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.  
**Assertion (A):**  $\text{I}_2$  is a mild oxidising agent.  
**Reason (R):**  $\text{I}_2$  can be used for titrating sodium thiosulphate.  
 In the light of the above statements, select the correct option.  
 (1) Both (A) and (R) are true and (R) is the correct explanation of (A)  
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)  
 (3) (A) is true but (R) is false  
 (4) (A) is false but (R) is true
88. Match List-I with List-II and select the correct option.

|    | List-I                                                                                                                                                          |       | List-II                         |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------|
| a. | $3\text{Mg}(\text{s}) + \text{N}_2(\text{g}) \xrightarrow{\Delta} \text{Mg}_3\text{N}_2(\text{s})$                                                              | (i)   | Disproportionation reaction     |
| b. | $\text{S}_8(\text{s}) + 12\text{OH}^-(\text{aq}) \rightarrow 4\text{S}^{2-}(\text{aq}) + 2\text{S}_2\text{O}_3^{2-}(\text{aq}) + 6\text{H}_2\text{O}(\text{l})$ | (ii)  | Non-metal displacement reaction |
| c. | $2\text{KClO}_3(\text{s}) \xrightarrow{\Delta} 2\text{KCl}(\text{s}) + 3\text{O}_2(\text{g})$                                                                   | (iii) | Decomposition reaction          |
| d. | $\text{Fe}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{FeCl}_2(\text{aq}) + \text{H}_2(\text{g})$                                                      | (iv)  | Combination reaction            |

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- (1) a(iv), b(i), c(iii), d(ii) (2) a(i), b(iv), c(ii), d(iii)  
 (3) a(iv), b(i), c(ii), d(iii) (4) a(i), b(iv), c(iii), d(ii)
89. Among the following solvents, solubility of AgCl will be minimum in  
 (1) 0.1 M AgNO<sub>3</sub> (2) 0.1 M CaCl<sub>2</sub>  
 (3) 0.5 M NaCl (4) 0.1 M NH<sub>3</sub>
90. If pK<sub>a</sub> of acid HX is 4, then K<sub>b</sub> of X<sup>-</sup> ion will be  
 (1)  $5 \times 10^{-9}$  (2)  $1 \times 10^{-10}$   
 (3)  $1 \times 10^{-4}$  (4)  $2 \times 10^{-5}$
91. CCl<sub>4</sub> cannot be hydrolysed because  
 (1) Carbon is less electronegative than chlorine  
 (2) Carbon has no vacant *d*-orbital  
 (3) Carbon has extremely small size  
 (4) CCl<sub>4</sub> is tetrahedral in shape
92. Formula unit present in pyrosilicates is  
 (1) SiO<sub>4</sub><sup>4-</sup> (2) Si<sub>4</sub>O<sub>6</sub><sup>4-</sup>  
 (3) Si<sub>2</sub>O<sub>7</sub><sup>6-</sup> (4) Si<sub>2</sub>O<sub>4</sub><sup>4-</sup>
93. Select an incorrect statement about borazine  
 (1) It has its molecular formula B<sub>3</sub>N<sub>3</sub>H<sub>6</sub>  
 (2) It is also known as inorganic benzene  
 (3) Nitrogen is sp<sup>3</sup> hybridized  
 (4) Boron is sp<sup>2</sup> hybridized
94. Consider the following statements.  
**Statement (I):** CO is highly poisonous in nature.  
**Statement (II):** CO prevents haemoglobin in the red blood corpuscles from carrying oxygen round the body.  
 Choose the **correct** option.  
 (1) Both statement (I) and statement (II) are correct  
 (2) Both Statements (I) and (II) are incorrect  
 (3) Statement (I) is correct but statement (II) is incorrect  
 (4) Statement (I) is incorrect but statement (II) is correct
95. Three centre-two electron bond is present in  
 (1) H<sub>3</sub>BO<sub>3</sub> (2) Na<sub>2</sub>B<sub>4</sub>O<sub>7</sub>  
 (3) B<sub>2</sub>H<sub>6</sub> (4) Al<sub>2</sub>Cl<sub>6</sub>
96. The pH of resulting solution obtained by mixing equal volume of 0.2 M HCl and 0.1 M H<sub>2</sub>SO<sub>4</sub> will be  
 (1) 0.7 (2) 1.2  
 (3) 0.4 (4) 1.5
97. The correct order of atomic radius of the given elements is  
 (1) In > Ga > Al > B (2) In > Al > Ga > B  
 (3) Ga > In > Al > B (4) Ga > Al > B > In
98. Which pair will not show common ion effect?  
 (1) CH<sub>3</sub>COOH + HCl  
 (2) HNO<sub>3</sub> + HCl  
 (3) NH<sub>4</sub>OH + NH<sub>4</sub>Cl  
 (4) HCOOH + HCOONa
99. Consider the following cell reaction.  
 $\text{Zn (s)} + \text{Ni}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Ni (s)}$   
 The value of E<sub>cell</sub><sup>°</sup> is (Given: E<sub>Ni<sup>2+</sup>/Ni</sub><sup>°</sup> = -0.25 V, E<sub>Zn<sup>2+</sup>/Zn</sub><sup>°</sup> = -0.76 V)  
 (1) 1.01 V (2) -1.01 V  
 (3) -0.51 V (4) 0.51 V
100. Consider the following statements.  
**Statement (I):** Aurous chloride and auric chloride are written as Au(I)Cl and Au(III)Cl<sub>3</sub> respectively.  
**Statement (II):** Hg<sub>2</sub>(l)Cl<sub>2</sub> is the oxidised form of Hg(II)Cl<sub>2</sub>.  
 Choose the **correct** option.  
 (1) Statement (I) is correct but statement (II) is incorrect  
 (2) Statement (I) is incorrect but statement (II) is correct  
 (3) Both Statements (I) and (II) are correct  
 (4) Both Statements (I) and (II) are incorrect

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**BOTANY****SECTION - A**

101. Which one of the following is not an inclusion body found in prokaryotes?
- (1) Cyanophycean granules
  - (2) Phosphate granules
  - (3) Glycogen granules
  - (4) Lysosome
102. The primary role of the contractile vacuole in *Amoeba* is
- (1) Photosynthesis
  - (2) Osmoregulation and excretion
  - (3) Protein synthesis
  - (4) Energy storage
103. Which process usually marks the end of the M-Phase?
- (1) DNA replication
  - (2) Centriole duplication
  - (3) Cytokinesis
  - (4) Karyokinesis
104. Match the following columns and select the **correct** option:

|    | Column I                                                     |       | Column II             |
|----|--------------------------------------------------------------|-------|-----------------------|
| a. | Synthesis of RNA and protein                                 | (i)   | S-phase               |
| b. | The centriole duplicates                                     | (ii)  | G <sub>0</sub> -phase |
| c. | Interval between mitosis and initiation of DNA replication   | (iii) | G <sub>1</sub> -phase |
| d. | The cells remain metabolically active but do not proliferate | (iv)  | G <sub>2</sub> -phase |

- (1) a(ii), b(iii), c(i), d(iv)    (2) a(iv), b(i), c(iii), d(ii)  
 (3) a(ii), b(i), c(iv), d(iii)    (4) a(ii), b(iv), c(i), d(iii)

105. Given below are two statements:

**Statement I:** The golgi apparatus modifies and packages materials for delivery within or outside the cell.

**Statement II:** Golgi apparatus is closely associate with the ER, in structural as well as functional aspects.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Statement I is correct but statement II is incorrect
  - (2) Statement I is incorrect but statement II is correct
  - (3) Both statements I and II are correct
  - (4) Both statements I and II are incorrect
106. Which among the following is a function of lysosomes in eukaryotic cells?
- (1) Energy production
  - (2) Synthesis of nucleic acid
  - (3) Digestion of macromolecules
  - (4) Involved in light dependent respiration
107. The enzyme recombinase is involved in which stage of prophase?
- (1) Leptotene                      (2) Pachytene
  - (3) Diplotene                    (4) Zygotene
108. Condensation and coiling of chromatin fibres begin during which sub-stage of prophase I?
- (1) Zygotene
  - (2) Leptotene
  - (3) Pachytene
  - (4) Diakinesis

Space for Rough Work

109. Which statement is **true** for ribosomes?
- (1) These are made up of ribosomal RNAs and proteins
  - (2) Free ribosomes synthesize lipids
  - (3) The prokaryotic ribosomes are 80S type
  - (4) These are composed of two identical subunits
110. What is the impact of meiosis on genetic variability in the population of organisms from one generation to the next?
- (1) It induces variation by reducing DNA content
  - (2) It involves recombination of sister chromatids and produces genetic recombinants
  - (3) It increases the chromosomal number
  - (4) It increases genetic variability
111. Which of the given components of mitochondria has oxysomes?
- (1) Cisternae                      (2) Grana
  - (3) Cristae                      (4) Centrosome
112. If a diploid cell with 20 chromosomes and 32 C DNA enters a cell cycle then what will be the number of chromosomes and DNA content after S-phase?
- (1) 20 chromosomes and 32 C DNA
  - (2) 20 chromosomes and 64 C DNA
  - (3) 40 chromosomes and 32 C DNA
  - (4) 40 chromosomes and 64 C DNA
113. Cytoskeleton are involved in all of the given functions, **except**
- (1) Maintenance of shape of cells
  - (2) Mechanical support to cell
  - (3)  $\beta$ -oxidation of fats
  - (4) Motility
114. Identify the correct statements and select the correct option
- a. Cytokinesis marks the end of cell division.
  - b. Cells exit the  $G_1$  phase to enter into  $G_0$  stage.
  - c. Mitochondria and plastids get doubled in  $G_2$  phase and distributed between daughter cells during cytokinesis.
  - d. Anaphase I involves separation of identical sister chromatids.
- (1) a and b only                      (2) b, c and d
  - (3) a, b and c                      (4) b and d
115. Plants grow continuously throughout their life because of the occurrence of mitotic divisions in
- A. Apical meristem              B. Lateral meristems
  - C. Parenchyma                  D. Collenchyma
- (1) Both A and B                  (2) Both A and C
  - (3) Both B and C                  (4) Both C and D
116. Which of the given is **not** true for the most dramatic period of cell cycle?
- (1) Reorganisation of all components of the cell
  - (2) Starts with the nuclear division
  - (3) Lasts for more than 95% of the duration of cell cycle
  - (4) It involves active cell division
117. Arrange the following in ascending order of their size
- P. Red blood cells
  - Q. Mycoplasma
  - R. Bacteria
- (1)  $Q < R < P$                       (2)  $Q < P < R$
  - (3)  $P < R < Q$                       (4)  $R < Q < P$
118. Select the events which occur twice during meiosis.
- a. Nuclear division
  - b. Division of cytoplasm
  - c. DNA replication
  - d. Bivalent formation
  - e. Equational division

Space for Rough Work

- (1) a and b only                      (2) a, b and e  
(3) b and d only                      (4) c, d and e
119. In which of the given stage of mitosis, the metacentric chromosomes clearly appear as V-shaped?
- (1) Prophase                              (2) Anaphase  
(3) Metaphase                            (4) Telophase
120. Kinetochores are
- (1) Terminal ends of chromosomes  
(2) Present around primary constriction and help in formation of bivalent  
(3) Present near nuclear envelope and take part in spindle formation  
(4) Present around centromere and provide a site for attachment to the spindle fibers
121. The three parts of a bacterial flagellum are
- (1) Filament, hook and basal body  
(2) Filament, basal body and axoneme  
(3) Hook, basal body and axoneme  
(4) Basal body, satellite and central hub
122. Select the **correct** statement(s) regarding the structure of a chromosome.
- a. Telomeres seal the ends of chromosome and prevent their shortening or chromosome loss.  
b. Two sister chromatids are held together by the centromere  
c. In sub-metacentric chromosome, one extremely short and one very long arm is present
- (1) a and c                              (2) b and c  
(3) Only a                                (4) a and b
123. Site of synthesis of ribosomal RNA is
- (1) Nuclear membrane    (2) Nucleolus  
(3) Lysosomes                      (4) Ribosomes
124. What will be the outcome if the pH in lysosomes is increased to alkaline?
- (1) The lysosomal enzymes will become more active  
(2) Hydrolytic enzymes will function more efficiently  
(3) Hydrolytic enzymes of lysosome become inactive  
(4) Carbohydrate, lipids, proteins will digest more easily
125. All of the following statements are true for metaphase, **except**
- (1) Condensation of chromosomes is completed  
(2) Spindle fibres attach to small disc-shaped structure of chromosomes  
(3) Chromosomes are moved to spindle equator and get aligned along metaphase plate through spindle fibres to both poles  
(4) Chromosomes cluster at opposite spindle poles and their identity is lost as discrete elements
126. Read the following Assertion (A) and Reason (R) and select the **correct** option.
- Assertion (A):** Meiosis introduces new combination of traits in the progeny.  
**Reason (R):** In Zygotene stage, homologous chromosomes pair to form bivalents, and undergo crossing over.
- (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)  
(2) (A) is true but (R) is false  
(3) Both (A) and (R) are false  
(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

Space for Rough Work

127. Read the following statements and select the **correct** option

**Statement A:** Chiasmata forms in diakinesis stage.

**Statement B:** Cells at the end of prophase, when viewed under the microscope, shows golgi complexes, ER, nucleolus and the nuclear envelope.

- (1) Only statement A is correct
- (2) Both statements A and B are incorrect
- (3) Both statements A and B are correct
- (4) Only statement B is correct

128. Which of the following is the **correct** sequence of occurrence of sub stages of prophase I?

- (1) Diakinesis → Diplotene → Pachytene → Zygotene → Leptotene
- (2) Leptotene → Zygotene → Pachytene → Diplotene → Diakinesis
- (3) Diplotene → Zygotene → Leptotene → Pachytene → Diakinesis
- (4) Leptotene → Diakinesis → Zygotene → Diplotene → Pachytene

129. Match the following columns and select the **correct** option.

|    | Column I                       |       | Column II             |
|----|--------------------------------|-------|-----------------------|
| a. | Ribosomes                      | (i)   | Schleiden and Schwann |
| b. | Golgi apparatus                | (ii)  | George Palade         |
| c. | <i>Omnis cellula-e cellula</i> | (iii) | Camillo Golgi         |
| d. | Cell theory                    | (iv)  | Rudolf Virchow        |

- (1) a(iii), b(ii), c(i), d(iv)
- (2) a(ii), b(iv), c(i), d(iii)
- (3) a(ii), b(iii), c(iv), d(i)
- (4) a(iv), b(i), c(ii), d(iii)

130. Read the following statements w.r.t. significance of mitosis

- a. The growth of multicellular organisms is due to mitosis.
- b. It helps in cell repair.
- c. It helps to restore the nucleo-cytoplasmic ratio of cell.
- d. It leads to variation which are very important for the process of evolution.

Select the **correct** option.

- (1) Only b is incorrect
- (2) a, b and d are correct
- (3) Only d is correct
- (4) a, b and c are correct

131. The start of second phase of mitosis is marked by

- (1) Formation of asters and spindle euqator
- (2) Complete disintegration of nuclear envelope
- (3) Movement of chromosomes to spindle equator
- (4) Splitting of centromere and separation of chromatids

132.  $\text{Na}^+ - \text{K}^+$  pump exemplifies

- (1) Active transport
- (2) Passive transport
- (3) Transport of molecules which does not use ATP
- (4) An energy independent process

133. In oocytes of some vertebrates, which of the given sub-stages of prophase-I can last for months or years?

- (1) Leptotene
- (2) Zygotene
- (3) Pachytene
- (4) Diplotene

134. Nuclear envelope is present in,

- (1) *Rhizobium*
- (2) *Ulothrix*
- (3) *Spirulina*
- (4) *Methanobacterium*

Space for Rough Work

135. The best stage to study the shape of the chromosomes is

- (1) Anaphase
- (2) Metaphase
- (3) Prophase
- (4) Telophase

### SECTION - B

136. Select the **incorrect** match

- |                                                     |                          |
|-----------------------------------------------------|--------------------------|
| (1) Membraneless structure                          | – Centrosome             |
| (2) Provide support to plasma membrane              | – Microfilament          |
| (3) They arise from the centriole like basal bodies | – Cilia and flagella     |
| (4) Pericentriolar satellites                       | – Form boundary of cilia |

137. Read the following Assertion (A) and Reason (R) statements and select the **correct** option.

**Assertion (A):** Mitochondria are called power house of the cell.

**Reason (R):** They produce cellular energy in the form of ATP.

- (1) (A) is true but (R) is false
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) Both (A) and (R) are false
- (4) Both (A) and (R) are true and (R) is the correct explanation of (A)

138. Read the following statements about plastids and state **true (T)** or **false (F)** for them

- a. Amyloplasts store carbohydrates.

b. In chromoplasts, fat soluble pigments like carotene and xanthophylls are present.

c. Aleuroplasts store oils and fats.

- |     | <b>a</b> | <b>b</b> | <b>c</b> |
|-----|----------|----------|----------|
| (1) | T        | F        | T        |
| (2) | T        | T        | F        |
| (3) | F        | T        | F        |
| (4) | F        | F        | T        |

139. At which stage chromosomal material gets untangled?

- (1) Metaphase
- (2) Prophase
- (3) Telophase
- (4) Anaphase

140. Which of the given is not true for a tetrad?

- (1) Two chromosomes having same genes on similar loci
- (2) Formed by a pair of chromosomes
- (3) Contains four sister chromatids
- (4) Clearly visible in pachytene stage

141. In plant cell, middle lamella is chiefly made up of

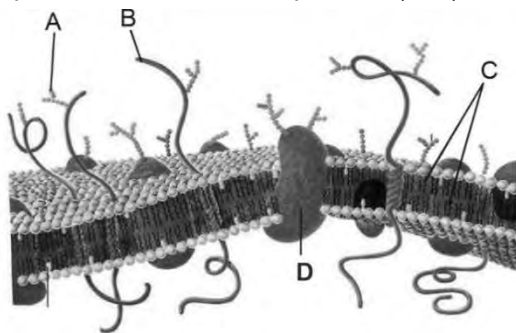
- (1) Cellulose
- (2) Mannans
- (3) Calcium pectate
- (4) Calcium carbonate

142. Select the **correct** option w.r.t. events which lead to formation of syncytium.

- (1) Karyokinesis not followed by cytokinesis
- (2) Karyokinesis not followed by disappearance of nuclear membrane
- (3) Nuclear division without DNA duplication
- (4) Cytokinesis followed by karyokinesis

Space for Rough Work

143. Consider the given diagram and select the **correct** option w.r.t. labelled components (A-D).



- |     | A           | B                  | C                    | D                  |
|-----|-------------|--------------------|----------------------|--------------------|
| (1) | Lipid       | Cholesterol        | Peripheral protein   | Integral protein   |
| (2) | Cholesterol | Integral protein   | Sugar                | Peripheral protein |
| (3) | Sugar       | Peripheral protein | Phospholipid bilayer | Integral protein   |
| (4) | Sugar       | Integral protein   | Cholesterol          | Lipid              |

144. Membrane proteins are classified into which of the following groups on the basis of ease of extraction?

- (1) Acidic and basic
- (2) Polar and non-polar
- (3) Integral and peripheral
- (4) Structural and enzymatic

145. Intracellular membranous compartmentalisation is the characteristic of

- (1) Prokaryotes
- (2) Viruses
- (3) Diatoms
- (4) *Nostoc*

146. If average duration of cell cycle of a normally dividing eukaryotic cell is 120 minutes then, what should be the duration of interphase in it?

- (1) 120 minutes
- (2) More than 114 minutes
- (3) 60 minutes
- (4) Less than 30 minutes

147. Mitosis is also known as equational division, because

- (1) Parent cell forms two daughter cells
- (2) Number of chromosomes in parent and progeny cells remains the same
- (3) There is an equal cytokinesis
- (4) Number of chromosomes becomes half in progeny as of the parent cell

148. If the gamete of an angiosperm has four chromosomes, then its parental cell had

- (1) 16 chromosomes during anaphase-I
- (2) 4 bivalents aligned on the equatorial plate during metaphase-I
- (3) 16 tetrads during pachytene
- (4) 16 chromosomes during G<sub>2</sub> stage

149. Which of the given events occur during S-phase of cell cycle?

- A. Replication of DNA
- B. Duplication of semi-autonomous cell organelles
- C. Synthesis of tubulin proteins
- D. Duplication of centrioles in the cytoplasm of higher plant cells.

- (1) A and D only
- (2) B and C only
- (3) A, C and D only
- (4) Only A

150. Select the **correct** sequence of events taking place in mitotic anaphase.

- A – Chromatids separate
- B – Centromeres split
- C – Chromatids move to opposite poles

- (1) B → A → C
- (2) C → B → A
- (3) A → B → C
- (4) A → C → B

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## ZOOLOGY

### SECTION - A

151. The outer pleural membrane is found in close proximity with all of the following, **except**
- (1) Diaphragm
  - (2) Ribs
  - (3) Sacrum
  - (4) Sternum
152. Rate of diffusion of  $O_2$  at alveoli and tissue site is independent of
- (1) Partial pressure of  $O_2$
  - (2) Solubility of  $O_2$
  - (3) Thickness of diffusion membrane
  - (4) Rate of diffusion of  $CO_2$
153. In a healthy adult man, under normal physiological conditions, the value of sum of TV, ERV and IRV is
- (1) Equal to FRC
  - (2) Less than TLC
  - (3) More than VC
  - (4) Equal to EC
154. Select the correct match w.r.t. organisms and their respiratory structures.
- |                         |   |                |
|-------------------------|---|----------------|
| (1) Prawn               | – | Gills          |
| (2) <i>Rana tigrina</i> | – | Tracheal tubes |
| (3) <i>Pterophyllum</i> | – | Lungs          |
| (4) <i>Struthio</i>     | – | Skin           |
155. All of the following are true w.r.t. enzyme carbonic anhydrase, **except**
- (1) Present in a very high concentration in RBCs
  - (2) At tissue site, it enables the formation of  $HCO_3^-$  and  $H^+$  from  $H_2CO_3$
  - (3) At alveolar site, it catalyses the formation of  $CO_2$  and  $H_2O$  from  $H_2CO_3$
  - (4) It's co-factor is  $Mg^{2+}$
156. The formation of oxyhaemoglobin at alveolar site is facilitated by all of the following factors, **except**
- (1) Low  $pCO_2$
  - (2) High  $pO_2$
  - (3) Low pH
  - (4) Low temperature
157. In the pulmonary vein, the partial pressure of gas which has 20-25 times higher solubility than that of  $O_2$  in blood is
- (1) Less than the partial pressure of  $CO_2$  in atmospheric air
  - (2) Equal to the partial pressure of  $O_2$  in tissues
  - (3) More than the partial pressure of  $O_2$  in atmospheric air
  - (4) More than the partial pressure of  $CO_2$  in pulmonary artery
158. **X** is the amount of  $O_2$  delivered to the tissues by every 300 mL of oxygenated blood whereas **Y** is the amount of  $CO_2$  delivered to the alveoli by every 400 mL of deoxygenated blood in an adult human, under normal physiological conditions. Identify the approximate value of **X** and **Y** and select the correct option.
- | X (mL) | Y (mL) |
|--------|--------|
| (1) 25 | 30     |
| (2) 5  | 4      |
| (3) 30 | 25     |
| (4) 15 | 16     |
159. Read the following statements carefully.
- Statement (A):** During forced expiration, internal inter-costal muscles relax leading to the pulling of ribs in inward direction.
- Statement (B):** During normal inspiration, diaphragm and external inter-costal muscles contract to increase the volume of thoracic chamber in the antero-posterior and dorso-ventral axis respectively.

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Select the **correct** option.

- (1) Both statements (A) and (B) are correct
- (2) Only statement (A) is correct
- (3) Both statements (A) and (B) are incorrect
- (4) Only statement (B) is correct

160. How many of the structures given in the box below constitute conducting part of human respiratory system which are also supported by incomplete cartilaginous rings?

Trachea, Primary bronchi, Tertiary bronchi, Initial bronchioles, Terminal bronchioles, Alveoli

Select the **correct** option.

- (1) Four                                      (2) Six
- (3) Three                                    (4) Two

161. *Erythroblastosis foetalis* can be avoided by administering \_\_\_\_\_ to the mother immediately after the delivery of the first child.

Select the **correct** option to fill in the blank.

- (1) Anti-Rh antigens
- (2) Anti-Rh antibodies
- (3) Rh toxoid
- (4) Rh antigens

162. In which of the following organisms, water does not circulate through its body cavity to facilitate the cells to exchange essential substances?

- (1) *Hydra*                                      (2) *Sycon*
- (3) *Obelia*                                    (4) *Pheretima*

163. In a cardiac cycle of a normal adult human, the flow of blood into the ventricles increases by about 30% during

- (1) Joint diastole
- (2) Atrial systole
- (3) Ventricular systole
- (4) Atrial diastole

164. Select the **incorrect** match w.r.t. a healthy man.

- (1) Water                                      – 90-92 per cent of blood plasma
- (2) Proteins                                    – 6-8 per cent of blood plasma
- (3) Formed elements                      – 45 per cent of blood plasma
- (4) Haemoglobin                            – 12-16 gms per decilitre of blood

165. During coagulation of blood, coagulum is formed mainly by a network of threads called

- (1) Fibrins
- (2) Thrombokinese
- (3) Thrombin
- (4) Thrombocytes

166. The oxygenated blood does not enter into the any heart chamber of

- (1) Amphibians
- (2) Reptiles
- (3) Mammals
- (4) Fishes

167. Though the whole of the nodal tissue is auto-excitabile but after removal of SAN, there would be great impact on the contractile activity of the heart. Which of the following options does not correctly justify the above scenario?

- (1) In nodal tissue, only SAN is responsible for the depolarisation of ventricles.
- (2) The SAN can generate maximum number of action potentials.
- (3) SAN is responsible for initiating the rhythmic contractile activity of the heart.
- (4) The SAN is the pacemaker of the heart.

Space for Rough Work

168. All of the following statements are true for lymph, **except**
- (1) It has same mineral distribution as that in plasma
  - (2) It is a colourless fluid containing specialised lymphocytes
  - (3) It contains larger proteins and most of the formed elements in it
  - (4) It is an important carrier for nutrients and hormones
169. Select the **incorrect** statement w.r.t. formed elements present in human blood.
- (1) RBCs are enucleated so as to provide all their internal space to haemoglobin for oxygen transport.
  - (2) Monocytes are agranulocytes which later transform into macrophages in tissues.
  - (3) Eosinophils are the cells responsible for secretion of histamine and serotonin.
  - (4) Basophils are the least abundant white blood cells.
170. One feature common to both fishes and humans is
- (1) Number of chambers present in their heart
  - (2) Presence of pulmonary circulation
  - (3) Only deoxygenated blood is pumped out by the heart
  - (4) Presence of closed type of circulation
171. A chronic disorder in which alveolar walls are damaged because of which respiratory surface is decreased and whose one of the major cause includes cigarette smoking is
- (1) Asthma
  - (2) Silicosis
  - (3) Pneumonia
  - (4) Emphysema
172. When percentage saturation of Hb with  $O_2$  is plotted against the  $pO_2$ , the shape of the curve will be
- (1) A straight line
  - (2) Sigmoid
  - (3) Hyperbola
  - (4) Parabola
173. The epithelial tissue present in the alveoli of human lungs is
- (1) Ciliated columnar
  - (2) Squamous
  - (3) Glandular
  - (4) Compound
174. The effect of sympathetic neural signals on the rate of heart beat is similar to the effect of \_\_\_\_\_ on heart.
- Select the **correct** option to fill in the blank.
- (1) ANF
  - (2) Norepinephrine
  - (3) Acetylcholine
  - (4) PTH
175. In a hypothetical scenario, what would be the immediate result, if the chordae tendineae connected to Mitral valve are cut?
- (1) Opening and closure of valve remains unaffected
  - (2) Decrease in the flow of blood in pulmonary artery
  - (3) Decrease in the flow of blood in aorta
  - (4) Increase in flow of blood in pulmonary veins
176. Choose the **correct** blood flow pathway in systemic circulation w.r.t. humans.
- (1) Right ventricle  $\xrightarrow{\text{Pulmonary artery}}$  Lungs  
 $\xrightarrow{\text{Pulmonary veins}}$  Left atrium
  - (2) Left ventricle  $\xrightarrow{\text{Aorta}}$  Body tissues  
 $\xrightarrow{\text{Vena cava}}$  Right atrium
  - (3) Right ventricle  $\xrightarrow{\text{Aorta}}$  Lungs  
 $\xrightarrow{\text{Pulmonary artery}}$  Left atrium
  - (4) Left ventricle  $\xrightarrow{\text{Pulmonary artery}}$  Body tissues  
 $\xrightarrow{\text{Vena cava}}$  Right atrium

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177. Select the conducting part of nodal tissue that passes through the proximal part of inter ventricular septum.
- (1) AV bundle
  - (2) Purkinje fibres
  - (3) Chordae tendineae
  - (4) Sino-atrial node
178. Upon holding the breath, the condition among the following which is mainly responsible for an urge to breath is
- (1) Falling  $pO_2$  in blood
  - (2) Increasing  $pCO_2$  in blood
  - (3) Increasing  $pO_2$  in blood
  - (4) Falling  $pCO_2$  in blood
179. Read the following statements carefully.
- Statement (A):** Tunica intima is the outer layer of wall of blood vessels made up of squamous endothelium.
- Statement (B):** The tunica media is comparatively thin in the veins than corresponding arteries.
- Select the **correct** option.
- (1) Both statements (A) and (B) are correct
  - (2) Both statement (A) and (B) are incorrect
  - (3) Only statement (A) is correct
  - (4) Only statement (B) is correct
180. In a hypothetical scenario, if the component of nodal tissue which is present in the lower left corner of the right atrium, close to the atrio ventricular septum is removed from human heart, what would be the effect of this removal?
- (1) The maintenance of the cardiac rhythmicity would be disturbed
  - (2) The contractile rhythmicity of the heart would remain normal as before
  - (3) The cardiac output would increase
  - (4) There would be delay in spreading the impulses to the atria of heart
181. Select the **incorrect** statement w.r.t. disorders of circulatory system.
- (1) In atherosclerosis, the coronary arteries get affected.
  - (2) Angina occurs when enough oxygen does not reach the heart muscles.
  - (3) Heart failure is the state of heart when it is suddenly damaged by an inadequate blood supply.
  - (4) Hypertension affects vital organs like brain and kidney.
182. 'X' are the cell fragments produced from megakaryocytes and reduction in their normal count can lead to clotting disorders.
- Identify 'X' and select its **correct** number per  $mm^3$  of human blood.
- (1) 5 millions - 5.5 millions
  - (2) 6000 - 8000
  - (3) 1.5 lakhs to 3.5 lakhs
  - (4) 30,000 - 60,000
183. Select the **incorrect** match
- |                 |   |                                            |
|-----------------|---|--------------------------------------------|
| (1) Fibrinogens | – | Required for coagulation of blood          |
| (2) Albumins    | – | Maintains osmotic balance in body          |
| (3) Globulins   | – | Involved in defense mechanisms of the body |
| (4) Serum       | – | Plasma with clotting factors               |
184. In a standard ECG, which of the following represents the **correct** activity w.r.t. human heart?
- (1) P-wave – Depolarisation of the atria
  - (2) QRS complex – Ending of the ventricular diastole
  - (3) T-wave – Depolarisation of the ventricles
  - (4) End of T-wave – Marks the beginning of ventricular systole

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185. The set of organisms in which blood is pumped out of the heart and the cells and tissues are directly bathed in it is

- (1) *Saccoglossus*, *Corvus*, *Ornithorhynchus*, *Culex*
- (2) *Anopheles*, *Locusta*, *Canis*, *Aplysia*
- (3) *Pheretima*, *Nereis*, *Bombyx*, *Apis*
- (4) *Pinctada*, *Limulus*, *Balanoglossus*, *Periplaneta*

### SECTION - B

186. Cardiac activity can be regulated by all, **except**

- (1) Neural centre present in medulla oblongata
- (2) Inputs of autonomic nervous system
- (3) Atrio-ventricular septum present in heart
- (4) Activity of adrenal medullary hormones

187. Suppose the cardiac output of a person is 5 L/min. Blood volume in left ventricle at the end of diastole and systole is 150 mL and 100 mL respectively. What would be his heart rate?

- (1) 100 beats per minute
- (2) 75 beats per minute
- (3) 120 beats per minute
- (4) 50 beats per minute

188. Read the following statements.

**Statement A:** The cardiac output of an athlete is usually higher than that of an ordinary man.

**Statement B:** The body has the ability to alter the stroke volume as well as the heart rate and thereby the cardiac output.

Choose the **correct** option.

- (1) Only statement A is correct
- (2) Only statement B is correct
- (3) Both statements A and B are correct
- (4) Both statements A and B are incorrect

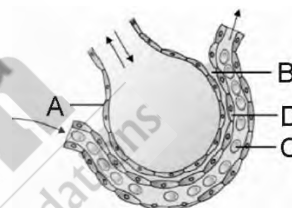
189. The middle layer of the walls of blood vessels in human body possesses

- (1) More white fibres and striated muscles
- (2) More yellow fibres and multinucleated muscle fibres
- (3) White fibres, muscle fibres and endothelium
- (4) Smooth muscles and elastic fibres

190. After the delivery of the first Rh +ve child, *erythroblastosis foetalis* could be a possibility for subsequent pregnancies in which of the following conditions?

|     | Blood group of mother | Blood group of foetus |
|-----|-----------------------|-----------------------|
| (1) | A <sup>+</sup>        | A <sup>-</sup>        |
| (2) | O <sup>-</sup>        | B <sup>+</sup>        |
| (3) | B <sup>+</sup>        | AB <sup>-</sup>       |
| (4) | O <sup>-</sup>        | O <sup>-</sup>        |

191. Observe the given figure.



Which of the following labelled structures is not the part of the diffusion membrane?

- (1) C
- (2) A
- (3) D
- (4) B

192. Higher number of RBCs in the blood of a mountain dweller in comparison to a person residing in plains can be explained by

- (1) Lack of pollutants in higher mountainous regions
- (2) Less  $pO_2$  in atmospheric air in mountains requiring more RBCs to supply the required amount of  $O_2$  to tissues

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- (3) Better nutrition intake by mountain dwellers  
 (4) Enhanced erythropoiesis in mountain dwellers due to high exposure to UV rays

193. **Assertion (A):** The two pleural membranes of human lungs fuse with each other during inhalation.

**Reason (R):** Fluid present between the two pleural membranes helps to maintain a positive pressure in the pleural cavity during inhalation.

In the light of above statements, choose the correct option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)  
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)  
 (3) (A) is true but (R) is false  
 (4) Both (A) and (R) are false

194. Which of the following statements is **incorrect** with respect to functional residual capacity?

- (1) Cannot be measured by a spirometer  
 (2) It includes ERV and RV  
 (3) It is the volume of air that will remain in the lungs after a normal expiration  
 (4) It is the volume of air remaining in the lungs even after a forcible expiration

195. Match the column I and column II w.r.t. transport of  $O_2$  and  $CO_2$  in humans.

|    | Column I            |       | Column II                            |
|----|---------------------|-------|--------------------------------------|
| a. | About 70% of $CO_2$ | (i)   | Transported as carbamino-haemoglobin |
| b. | About 97% of $O_2$  | (ii)  | Transported as bicarbonates          |
| c. | About 7% of $CO_2$  | (iii) | Transported as oxyhaemoglobin        |

|    |                         |      |                                               |
|----|-------------------------|------|-----------------------------------------------|
| d. | Nearly 20-25% of $CO_2$ | (iv) | Transported in dissolved state through plasma |
|----|-------------------------|------|-----------------------------------------------|

Select the **correct** option.

- (1) a(iv), b(ii), c(i), d(iii) (2) a(i), b(ii), c(iii), d(iv)  
 (3) a(ii), b(iii), c(iv), d(i) (4) a(iv), b(iii), c(ii), d(i)

196. Increased concentration of  $CO_2$  and hydrogen ions are being recognised by '**X**' situated adjacent to '**Y**' present in the human brain.

Select the **correct** option that identify 'X' and 'Y' respectively

- (1) Respiratory rhythm centre, Pneumotaxic centre  
 (2) Pneumotaxic centre, Chemosensitive area  
 (3) Chemosensitive area, Respiratory rhythm centre  
 (4) Chemosensitive area, Pneumotaxic centre

197. Consider this hypothetical scenario, where the cardiac output of a person is 7000 mL and his stroke volume is 70 mL.

Calculate the duration of his cardiac cycle.

- (1) 0.7 seconds  
 (2) 0.8 seconds  
 (3) 0.9 seconds  
 (4) 0.6 seconds

198. A person with anti-B antibodies in his plasma can donate blood to all, **except**

- (1) A person with both A and B antigens on his RBCs  
 (2) A person with no antibodies against blood antigens in his plasma  
 (3) A person with both anti-A and anti-B antibodies in his plasma  
 (4) A person with A antigens on his RBCs

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199. Match column I with column II.

|    | Column I                     |       | Column II                                    |
|----|------------------------------|-------|----------------------------------------------|
| a. | Opening of Mitral valve      | (i)   | During ventricular systole                   |
| b. | Opening of semilunar valves  | (ii)  | Between right ventricle and pulmonary artery |
| c. | Location of tricuspid valve  | (iii) | During ventricular diastole                  |
| d. | Location of semilunar valves | (iv)  | Between right atrium and right ventricle     |

Select the **correct** option.

- (1) a(i), b(ii), c(iii), d(iv) (2) a(iii), b(i), c(iv), d(ii)  
 (3) a(ii), b(i), c(iii), d(iv) (4) a(iv), b(i), c(ii), d(iii)

200. One of the significance of hepatic portal system is

- (1) To decrease the volume of blood reaching systemic circulation  
 (2) To directly utilize the nutrients and send them to pulmonary circulation  
 (3) That the absorbed nutrients directly reach the pulmonary circulation at a faster rate  
 (4) To eliminate the toxic substances present in absorbed food

□ □ □

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**Coping Mechanism for Stress Tips from our NEET Toppers**



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## FINAL TEST SERIES for NEET-2024

MM : 720

Test - 3

Time : 3 Hrs. 20 Mins.

### Answers

|         |         |          |          |          |
|---------|---------|----------|----------|----------|
| 1. (1)  | 41. (2) | 81. (3)  | 121. (1) | 161. (2) |
| 2. (2)  | 42. (2) | 82. (1)  | 122. (4) | 162. (4) |
| 3. (4)  | 43. (3) | 83. (1)  | 123. (2) | 163. (2) |
| 4. (1)  | 44. (2) | 84. (2)  | 124. (3) | 164. (3) |
| 5. (4)  | 45. (1) | 85. (3)  | 125. (4) | 165. (1) |
| 6. (3)  | 46. (2) | 86. (3)  | 126. (2) | 166. (4) |
| 7. (4)  | 47. (4) | 87. (2)  | 127. (2) | 167. (1) |
| 8. (3)  | 48. (2) | 88. (1)  | 128. (2) | 168. (3) |
| 9. (3)  | 49. (3) | 89. (3)  | 129. (3) | 169. (3) |
| 10. (1) | 50. (3) | 90. (2)  | 130. (4) | 170. (4) |
| 11. (2) | 51. (3) | 91. (2)  | 131. (2) | 171. (4) |
| 12. (4) | 52. (2) | 92. (3)  | 132. (1) | 172. (2) |
| 13. (4) | 53. (1) | 93. (3)  | 133. (4) | 173. (2) |
| 14. (4) | 54. (3) | 94. (1)  | 134. (2) | 174. (2) |
| 15. (2) | 55. (1) | 95. (3)  | 135. (1) | 175. (3) |
| 16. (3) | 56. (1) | 96. (1)  | 136. (4) | 176. (2) |
| 17. (1) | 57. (1) | 97. (2)  | 137. (4) | 177. (1) |
| 18. (3) | 58. (3) | 98. (2)  | 138. (2) | 178. (2) |
| 19. (3) | 59. (3) | 99. (4)  | 139. (2) | 179. (4) |
| 20. (1) | 60. (1) | 100. (1) | 140. (3) | 180. (1) |
| 21. (1) | 61. (3) | 101. (4) | 141. (3) | 181. (3) |
| 22. (3) | 62. (2) | 102. (2) | 142. (1) | 182. (3) |
| 23. (1) | 63. (1) | 103. (3) | 143. (3) | 183. (4) |
| 24. (3) | 64. (4) | 104. (2) | 144. (3) | 184. (1) |
| 25. (3) | 65. (2) | 105. (3) | 145. (3) | 185. (4) |
| 26. (1) | 66. (1) | 106. (3) | 146. (2) | 186. (3) |
| 27. (2) | 67. (2) | 107. (2) | 147. (2) | 187. (1) |
| 28. (2) | 68. (3) | 108. (2) | 148. (2) | 188. (3) |
| 29. (3) | 69. (2) | 109. (1) | 149. (4) | 189. (4) |
| 30. (2) | 70. (3) | 110. (4) | 150. (1) | 190. (2) |
| 31. (2) | 71. (1) | 111. (3) | 151. (3) | 191. (1) |
| 32. (4) | 72. (2) | 112. (2) | 152. (4) | 192. (2) |
| 33. (4) | 73. (2) | 113. (3) | 153. (2) | 193. (4) |
| 34. (2) | 74. (4) | 114. (3) | 154. (1) | 194. (4) |
| 35. (1) | 75. (1) | 115. (1) | 155. (4) | 195. (3) |
| 36. (2) | 76. (1) | 116. (3) | 156. (3) | 196. (3) |
| 37. (1) | 77. (1) | 117. (1) | 157. (2) | 197. (4) |
| 38. (1) | 78. (4) | 118. (1) | 158. (4) | 198. (3) |
| 39. (1) | 79. (3) | 119. (2) | 159. (4) | 199. (2) |
| 40. (1) | 80. (3) | 120. (4) | 160. (1) | 200. (4) |



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## FINAL TEST SERIES for NEET-2024

MM : 720

Test - 3

Time : 3 Hrs. 20 Mins.

### Answers and Solutions

#### PHYSICS

##### SECTION - A

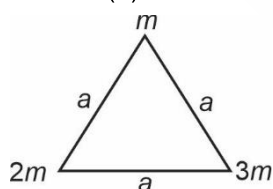
1. Answer (1)

$$g = \frac{Gm}{r^2}$$

$$\frac{g}{G} = \frac{m}{r^2}$$

Hence, unit of  $\frac{g}{G}$  is  $\frac{\text{kg}}{\text{m}^2}$

2. Answer (2)



$$U = \frac{-2Gm^2}{a} - \frac{6Gm^2}{a} - \frac{3Gm^2}{a}$$

$$= \frac{-11Gm^2}{a}$$

3. Answer (4)

$$T.E = -E$$

$$B.E = E$$

4. Answer (1)

As per Kepler's law of time period

$$T^2 \propto R^3$$

5. Answer (4)

$$V \times \rho_b g = V_{in} \times \rho_\ell g$$

$$\frac{\rho_b}{\rho_w} = \left( \frac{V_{in}}{V} \right)$$

$$= \frac{70}{100}$$

$$= 0.7$$

6. Answer (3)

$$P_B = P_A + \rho g_{eff} h$$

$$P_B = P_A + \rho \frac{4gh}{3}$$

$$P_B - P_A = \left( \frac{4\rho gh}{3} \right)$$

7. Answer (4)

Water droplets are spherical in free fall condition due to surface tension.

8. Answer (3)

$$\Delta \ell = \frac{F\ell}{YA}$$

$$\Delta \ell \propto \ell \quad (F, Y \text{ and } A \text{ are same})$$

$$\frac{\Delta \ell_1}{\Delta \ell_2} = \frac{\ell_1}{\ell_2} = \frac{2}{3}$$

9. Answer (3)

Longitudinal strain is meaningful in solid.

10. Answer (1)

Steel is more elastic than rubber and hooke's law is applicable up to elastic limit.

11. Answer (2)

$$\frac{K-273}{100} = \frac{F-32}{180}$$

$$K-273 = \frac{5}{9}(F-32)$$

$$K_1 = \frac{5}{9}(F_1-32) + 273$$

$$K_2 = \frac{5}{9}(F_2-32) + 273$$

$$K_2 - K_1 = \frac{5}{9}(F_2 - F_1)$$

$$50 = \frac{5}{9}n$$

$$n = 90$$

12. Answer (4)

$$\frac{130-50}{R} + \frac{20-50}{R} + \frac{x-50}{R} = 0$$

$$x = 0^\circ\text{C}$$

13. Answer (4)

$$\begin{aligned}\Delta P &= \frac{2T}{R} \\ &= \frac{2 \times 0.072}{1.2 \times 10^{-3}} \\ &= \frac{2 \times 7.2 \times 10^{-2}}{1.2 \times 10^{-3}} \\ &= 2 \times 6 \times 10 \\ &= 120 \text{ N/m}^2\end{aligned}$$

14. Answer (4)

$$h = \frac{2s \cos \theta}{\rho g r}$$

$$\frac{h_1}{h_2} = \frac{r_2}{r_1} = 2 : 1$$

15. Answer (2)

The working of venturimeter is based on Bernoulli's theorem.

16. Answer (3)

$$R = 2\sqrt{h(H-h)}$$

$$\text{For } R = R_{\max}$$

$$\frac{dR}{dh} = 0$$

$$h = \frac{H}{2}$$

17. Answer (1)

From mechanical energy conservation

$$\frac{1}{2}mv^2 - \frac{3GMm}{2R} = 0 \Rightarrow v = \sqrt{\frac{3GM}{R}} = \sqrt{1.5} v_e$$

18. Answer (3)

$$F_1 = F_2 = \frac{G(1)(1)}{(2)^2} = \frac{G}{4}$$

$$\vec{F}_{\text{net}} = F_1(\hat{i}) + F_2(\hat{j}) = F(\hat{i} + \hat{j}) = \frac{G}{4}(\hat{i} + \hat{j}) \text{ N}$$

19. Answer (3)

As the height decreases, the rate of flow with which the water is coming out decreases.

20. Answer (1)

As the cork moves up, the force due to buoyancy remains constant. As its speed increases, the retarding force due to viscosity increases, being proportional to the speed. Thus the acceleration gradually decreases.

21. Answer (1)

$$\text{Velocity of efflux} = \sqrt{2gh} = \sqrt{2 \times 10 \times 5} = 10 \text{ m/s}$$

$$\text{Rate of flow} = Av = (1 \times 10^{-4}) \times 10 = 10^{-3} \text{ m}^3/\text{s}$$

22. Answer (3)

$$r_{\text{new}} = \sqrt{r_1^2 + r_2^2} = \sqrt{3^2 + 4^2} = 5 \text{ cm}$$

23. Answer (1)

$$t = \frac{(P_t - P_0)}{(P_{100} - P_0)} \times 100^\circ\text{C} = \frac{(60 - 50)}{(90 - 50)} \times 100 = 25^\circ\text{C}$$

24. Answer (3)

When the temperature of a liquid is increased by  $\Delta T^\circ\text{C}$  the mass will remain unchanged while due to thermal expansion volume will increase and become  $V'$ .

$$V' = V(1 + \gamma\Delta T)$$

where  $\gamma$  is the coefficient of volume expansion of liquid.

$$\rho' = \frac{m}{V'} = \frac{m}{V(1 + \gamma\Delta T)} = \frac{\rho}{1 + \gamma\Delta T} \quad \left[ \because \rho = \frac{m}{V} \right]$$

$$\text{Fractional change in density} = \frac{\rho - \rho'}{\rho} = \frac{\Delta\rho}{\rho} = \frac{\delta v}{v}$$

$$= \gamma\Delta T$$

$$= 49 \times 10^{-5} \times 30$$

$$= 1.5 \times 10^{-2}$$

25. Answer (3)

The densest layer of water will be at bottom. The density of water is maximum at  $4^\circ\text{C}$ . So the temperature of bottom of lake will be  $4^\circ\text{C}$ .

26. Answer (1)

$$R = R_1 + R_2$$

$$\frac{2\ell}{k_{\text{eff}}A} = \frac{\ell}{kA} + \frac{\ell}{3kA}$$

$$\frac{2\ell}{k_{\text{eff}}} = \frac{\ell}{k} + \frac{\ell}{3k}$$

$$\frac{2}{k_{\text{eff}}} = \frac{4}{3k}$$

$$K_{\text{eff}} = \frac{3k}{2}$$

27. Answer (2)

$$\Delta\ell = \frac{FL}{AY} \Rightarrow \Delta\ell \propto \frac{L}{r^2}$$

28. Answer (2)

By Bernoulli's theorem,  $\frac{1}{2}\rho v_1^2 + P_1 = \frac{1}{2}\rho v_2^2 + P_2$

$$P_1 - P_2 = \frac{1}{2}\rho v_2^2 - \frac{1}{2}\rho v_1^2$$

$$2h\rho g = (v_2^2 - v_1^2)\rho$$

$$h = \frac{v_2^2 - v_1^2}{2g} = 0.8 \text{ m}$$

29. Answer (3)

$$(l_1 - l_0) = \frac{T_1 l_0}{AY} \quad \dots \text{eq. (1)}$$

$$(l_2 - l_0) = \frac{T_2 l_0}{AY} \quad \dots \text{eq. (2)}$$

Equation (1) divided by (2)

$$\frac{l_1 - l_0}{l_2 - l_0} = \frac{T_1}{T_2}$$

$$T_2 l_1 - T_2 l_0 = T_1 l_2 - T_1 l_0$$

$$T_2 l_0 - T_1 l_0 = T_2 l_1 - T_1 l_2$$

$$l_0 (T_2 - T_1) = T_2 l_1 - T_1 l_2$$

$$l_0 = \left( \frac{T_2 l_1 - T_1 l_2}{T_2 - T_1} \right)$$

30. Answer (2)

The Young's modulus of the material of a wire is equal to the stress required to produce unit strain.

31. Answer (2)

$$g' = \frac{GM}{r^2} = \frac{GM}{(R+h)^2}$$

$$\text{Now, } g' = \frac{g}{9}$$

$$\Rightarrow \frac{GM}{(R+h)^2} = \frac{1}{9} \frac{GM}{R^2}$$

$$\Rightarrow h = 2R$$

32. Answer (4)

$$T^2 \propto r^3$$

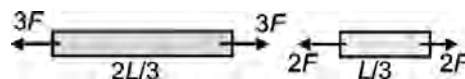
$$\left( \frac{84}{T} \right)^2 = \left( \frac{r}{3r} \right)^3$$

$$\Rightarrow T = 3\sqrt{3} \times 84$$

$$= 435.96$$

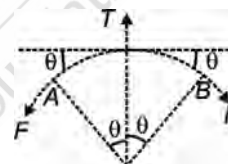
$$\approx 436 \text{ days}$$

33. Answer (4)



$$I = \frac{3F \left( \frac{2L}{3} \right)}{AY} + \frac{2F \left( \frac{L}{3} \right)}{AY} \Rightarrow I = \frac{8FL}{3AY}$$

34. Answer (2)



$F$  = Tension in the thread

$$2F \sin\theta = 2T(AB) \quad (\because \sin\theta = \frac{R}{2R})$$

$$2F \sin\theta = 2T(2R\sin\theta)$$

$$F = 2RT$$

35. Answer (1)

$$v_e = \sqrt{2gR}, \text{ Velocity at infinity is zero.}$$

### SECTION-B

36. Answer (2)

According to Kepler's law of areal velocity

$$\frac{dA}{dt} = \frac{L}{2m}$$

37. Answer (1)

At any point below the surface of earth

$$g = \frac{GM}{R^3} r$$

$$g \propto r$$

38. Answer (1)

Potential energy at the surface of the earth

$$= -\frac{GMm}{R}$$

$$= -gmR$$

Potential energy at a distance  $R$  from the surface

$$\text{of earth} = -\frac{GMm}{R+R}$$

$$= -\frac{GMm}{2R}$$

$$= -\frac{mgR}{2}$$

$$\text{Gain in potential energy} = -\frac{gmR}{2} - (-gmR)$$

$$= \frac{gmR}{2}$$

39. Answer (1)

Since the escape velocity is independent of direction of projection, therefore escape velocity of body, projected at an angle  $45^\circ$  with the vertical will be same i.e. 11 km/s.

40. Answer (1)

$$\frac{\Delta \ell}{\ell} = \left( \frac{1}{100} \right)$$

$$V = \ell^3$$

$$\frac{\Delta V}{V} = 3 \frac{\Delta \ell}{\ell}$$

$$= 3 \times (0.01)$$

$$= 0.03$$

41. Answer (2)

$$a + r + t = 1$$

$$0.2 + 0.4 + t = 1$$

$$t = 0.4$$

$$Q = 400 \times 0.4 \times 10$$

$$= 1600 \text{ J}$$

42. Answer (2)

$$m_1 S_1 \Delta T = m_2 S_2 \Delta T$$

$$m_1 \times S_1 \times 4 = m_2 \times S_2 \times 6$$

$$\frac{S_1}{S_2} = \frac{m_2}{m_1} \times \frac{3}{2}$$

$$= \frac{1}{2} \times \frac{3}{2}$$

$$= \frac{3}{4}$$

43. Answer (3)

Terminal velocity of a spherical body in viscous fluid is

$$V_T = \frac{2}{9\eta} r^2 (\rho_s - \rho_l)$$

$$V_T \propto r^2$$

44. Answer (2)

Height of liquid column

$$h = \frac{2T \cos \theta}{r \rho g}$$

$$\text{If } h = 0$$

$$\therefore \theta = 90^\circ$$

45. Answer (1)

$$1000r^3 = R^3$$

$$r = \frac{R}{10}$$

$$A_1 = 4\pi r^2$$

$$A_2 = 1000 \times 4\pi r^2 = 40\pi R^2$$

$$\text{Change in energy} = (\Delta A)T$$

$$= 36\pi R^2 T$$

$$= 36\pi \times 10^{-4} \times 0.075$$

$$= 27\pi \times 10^{-5} \text{ J}$$

46. Answer (2)

$$\frac{v^2}{2g} = 10 \text{ cm}$$

$$\frac{v^2}{2 \times 10} = 10 \times 10^{-2} \text{ m}$$

$$v = \sqrt{2 \times 10^2 \times 10^{-2}}$$

$$v = \sqrt{2} \text{ m/s}$$

$$\simeq 1.4 \text{ m/s}$$

47. Answer (4)

$$\tan 37^\circ = Y_B$$

$$\tan 53^\circ = Y_A$$

$$\frac{Y_A}{Y_B} = \frac{\frac{4}{3}}{\frac{3}{4}} = \frac{16}{9}$$

$$Y_B = \frac{9}{16} Y_A$$

48. Answer (2)

The strip will bend having concave surface on side 1.

49. Answer (3)

Water will expand on heating as well as on cooling from 4°C.

50. Answer (3)

$$\omega^2 = \frac{GM}{r^3}$$

$$(\omega')^2 = \frac{GM}{(4r)^3}$$

$$\frac{\omega^2}{(\omega')^2} = \left(\frac{4r}{r}\right)^3 = 64$$

$$\frac{\omega}{\omega'} = 8$$

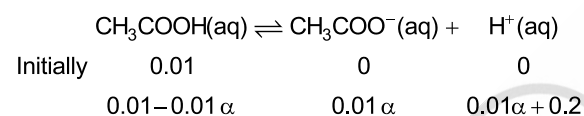
## CHEMISTRY

### SECTION - A

51. Answer (3)

Salt of strong acid and weak base  $[(\text{NH}_4)_2\text{SO}_4]$  on hydrolysis form acidic solution.

52. Answer (2)



$$K_a = \frac{0.01\alpha(0.01\alpha + 0.2)}{0.01(1-\alpha)} \alpha \lll 1$$

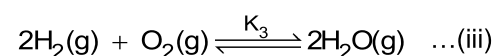
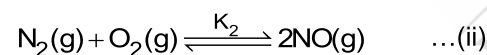
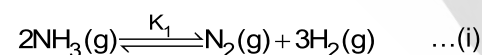
$$K_a = 0.2\alpha$$

$$\alpha = \frac{1.8 \times 10^{-5}}{0.2}$$

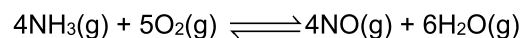
$$= 9 \times 10^{-5}$$

$$\therefore \% \text{ ionization} = 0.009\%$$

53. Answer (1)

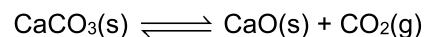


Applying  $2 \times (\text{i}) + 2 \times (\text{ii}) + 3 \times (\text{iii})$ , we get



$$K = K_1^2 K_2^2 K_3^3$$

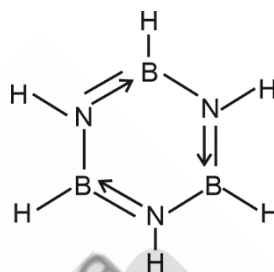
54. Answer (3)



$$\Delta n_g = (1) - (0) = 1$$

$$\therefore K_P = K_C(\text{RT})^{\Delta n_g} = K_C \text{RT}$$

55. Answer (1)



Number of sigma bonds = 12

Number of pi bonds = 3

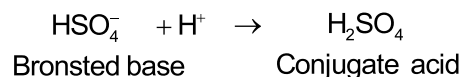
56. Answer (1)

On increasing temperature, dissociation of water increases means  $K_w$  increases but pH and pOH both decreases.

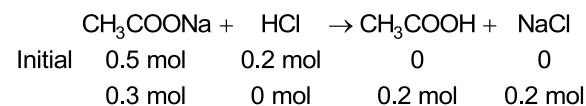
57. Answer (1)

$$K_C = \frac{[\text{D}_2]^3}{[\text{A}_2][\text{B}_2]^2}$$

58. Answer (3)



59. Answer (3)

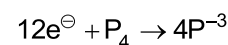


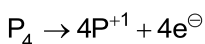
$$\text{Now, } \text{pH} = \text{p}K_a + \log \frac{[\text{Salt}]}{[\text{Acid}]}$$

$$= 4.74 + \log \frac{0.3}{0.2}$$

$$= 4.74 + \log \frac{3}{2} = 4.92$$

60. Answer (1)





$\therefore$  4 mole  $P_4 \Rightarrow$  12 mole  $e^{\ominus}$  gain or loss

$\therefore$  1 mole  $P_4 \Rightarrow \frac{12}{4} = 3$  mole  $e^{\ominus}$

$$\text{Equivalent weight} = \frac{M}{3}$$

61. Answer (3)

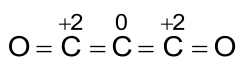
In  $H_2S$ , S is present in its lowest oxidation state *i.e.*,  $-2$ , so it cannot be further reduced.

62. Answer (2)

Balance reaction is



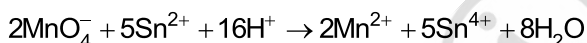
63. Answer (1)



64. Answer (4)

In  $HClO_4$ , Cl is at its maximum oxidation state *i.e.*,  $+7$ .

65. Answer (2)



66. Answer (1)

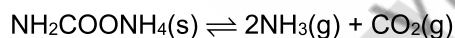
$MnO_4^-$  can act as a self-indicator.

67. Answer (2)

Reduction potential  $K < Zn < Cr < Ag$ .

Therefore, Zn can reduce  $Cr^{3+}$  and  $Ag^+$  but not  $K^+$ .

68. Answer (3)

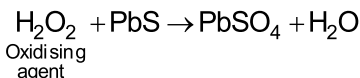
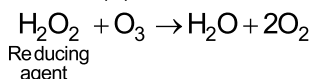


(i)  $2P$   $P$   
(f)  $P'$   $(2P)$

$$\therefore K_P = (P')^2(2P) = (2P)^2(P)$$

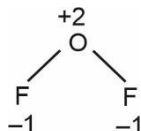
$$\Rightarrow P' = \sqrt{2}P = \frac{\sqrt{2}}{3} \times 3P = \frac{\sqrt{2}}{3} \times (P_{\text{total}})$$

69. Answer (2)



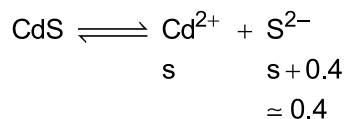
70. Answer (3)

In  $OF_2$ , the oxygen is assigned an oxidation number of  $+2$ .



71. Answer (1)

Let the solubility of  $CdS$  be  $s$  mol/L



$$K_{sp} = [Cd^{2+}][S^{2-}]$$

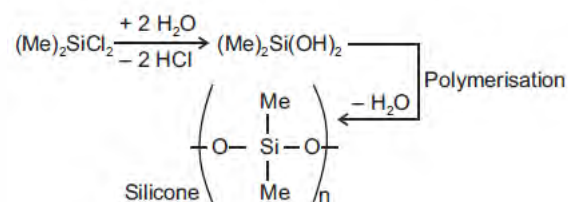
$$8 \times 10^{-27} = s(0.4)$$

$$s = 2 \times 10^{-26}$$

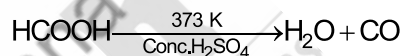
72. Answer (2)

Addition of  $N_2(g)$  at equilibrium will shift the reaction in forward direction.

73. Answer (2)



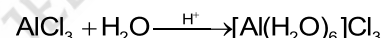
74. Answer (4)



75. Answer (1)

In  $H_3PO_4$ , it is more difficult to remove a proton from a doubly charged  $HPO_4^{2-}$  anion as compared to  $H_2PO_4^-$ .

76. Answer (1)



Complex A is  $[Al(H_2O)_6]^{3+}$  and B is  $sp^3d^2$

77. Answer (1)

CO : Neutral

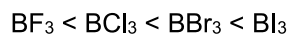
$In_2O_3$  : Basic

$Ga_2O_3$  : Amphoteric

$SiO_2$  : Acidic

78. Answer (4)

Order of Lewis acid nature:



79. Answer (3)

Graphite has layered structure with  $sp^2$  hybridised carbon.

80. Answer (3)

$CaOCl_2$  exists as  $Ca^{2+}$ ,  $Cl^-$  and  $OCl^-$ .

$\therefore$  O.S of chlorine in  $CaOCl_2$  is  $-1$  and  $+1$ .

81. Answer (3)

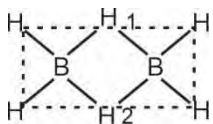
Solution of salt of weak acid and weak base acts as a buffer.

82. Answer (1)

At equilibrium,  $\Delta G^\circ = -RT \ln K$

$$\begin{aligned}\Delta G^\circ &= -2.303 \times 2 \times 300 \times \log 10^3 \\ &= -2.303 \times 2 \times 300 \times 3 \log 10 \\ &= -2.303 \times 1800 \\ &= -4.14 \text{ kcal mol}^{-1}\end{aligned}$$

83. Answer (1)



Contain 2 B-H-B bonds.

84. Answer (2)

$\text{Pb}^{2+}$  is more stable than  $\text{Pb}^{4+}$  (inert pair effect).

85. Answer (3)

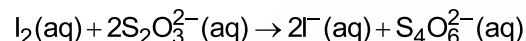
When air is used instead of steam, a mixture of CO and  $\text{N}_2$  is produced, which is called producer gas.

### SECTION-B

86. Answer (3)

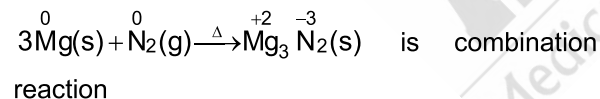
On adding  $\text{CO(g)}$ ,  $\text{CoCl}_2(\text{g})$  will increase and  $\text{PCl}_5(\text{g})$  will decrease.

87. Answer (2)



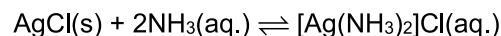
Iodine itself gives an intense blue colour with starch and has a very specific reaction with  $\text{S}_2\text{O}_3^{2-}$ .

88. Answer (1)



89. Answer (3)

Due to complex formation solubility of AgCl will be maximum in 0.1 M  $\text{NH}_3$  and minimum in 0.5 M NaCl.



90. Answer (2)

$$\text{pK}_a = -\log K_a$$

For conjugate acid-base pair

$$K_a \times K_b = K_w$$

$$K_b = \frac{K_w}{K_a}$$

$$= \frac{10^{-14}}{1 \times 10^{-4}}$$

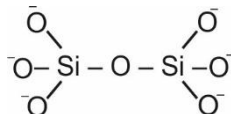
$$= 1 \times 10^{-10}$$

91. Answer (2)

$\text{CCl}_4$  cannot get hydrolysed due to absence of empty  $d$ -orbital in outermost shell of carbon atom.

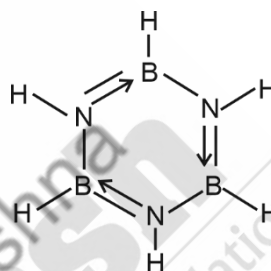
92. Answer (3)

Pyrosilicates have  $\text{Si}_2\text{O}_7^{6-}$  unit



93. Answer (3)

Borazine is also known as inorganic benzene



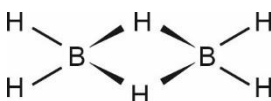
Both B and N are  $sp^2$  hybridised.

94. Answer (1)

Highly poisonous nature of CO arises because of its ability to form a complex with haemoglobin, which is about 300 times more stable than oxygen-haemoglobin complex.

95. Answer (3)

$\text{B}_2\text{H}_6$  contains  $3\text{C} - 2\text{e}$  bond



96. Answer (1)

$$[\text{H}^+] = \frac{0.2 \times V + 0.1 \times 2 \times V}{V + V} = 0.2 \text{ M}$$

$$\therefore \text{pH} = -\log(0.2) = 0.7$$

97. Answer (2)

| Element | Atomic radius (pm) |
|---------|--------------------|
| B       | 88                 |
| Al      | 143                |
| Ga      | 135                |
| In      | 167                |

98. Answer (2)

$\text{HNO}_3 + \text{HCl}$  will not show common ion effect due to strong acidic nature of  $\text{HNO}_3$  and  $\text{HCl}$ .

99. Answer (4)

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$$

$$= -0.25 - (-0.76)$$

$$= 0.51 \text{ V}$$

100. Answer (1)

- $\text{Hg}_2(\text{I})\text{Cl}_2$  is the reduced form of  $\text{Hg}(\text{II})\text{Cl}_2$ .
- Aurous chloride and Auric chloride are written as  $\text{Au}(\text{I})\text{Cl}$  and  $\text{Au}(\text{III})\text{Cl}_3$  respectively.

## BOTANY

### SECTION - A

101. Answer (4)

Lysosomes are membrane bound vesicular structure formed by the process of packaging in the golgi apparatus.

102. Answer (2)

In *Amoeba*, the contractile vacuole is important for osmoregulation and excretion.

103. Answer (3)

The M phase usually ends with division of cytoplasm *i.e.*, cytokinesis.

104. Answer (2)

$G_1$  phase is the interval between mitosis and initiation of DNA replication.

In  $G_0$  phase cell remains metabolically active but do not proliferate unless called on to do so.

In the amount of nucleic acid (DNA) doubles in S-phase.

105. Answer (3)

The important function of Golgi apparatus is to process, package and transport the materials for secretions. It is closely associated with ER in structural as well as functional aspects.

106. Answer (3)

Lysosomes are rich in hydrolytic enzymes that are capable of digesting carbohydrates, proteins, lipids and nucleic acid.

107. Answer (2)

During pachytene, crossing over occurs between non-sister chromatids of the homologous chromosomes which is an enzyme mediated process and the enzyme involved is called recombinase.

108. Answer (2)

Condensation and coiling of chromatin fibres begin during leptotene.

109. Answer (1)

Ribosomes, made up of ribosomal RNAs and proteins, are the cellular components that carryout protein synthesis.

110. Answer (4)

Meiosis increases the genetic variability in the population of organisms from one generation to the next.

111. Answer (3)

Each oxysome is differentiated into base, stalk and headpiece. The headpiece contains ATP synthetase which brings about oxidative phosphorylation coupled with release of ATP.

112. Answer (2)

In S-phase, the amount of DNA doubles but number of chromosomes remain same.

Cell with 20 chromosomes and 32 C DNA in  $G_1$  phase will have 20 chromosomes and 64 C DNA after S-phase.

113. Answer (3)

$\beta$ -oxidation of fats occurs in mitochondria.

114. Answer (3)

Sister chromatids separate during Anaphase II.

115. Answer (1)

Mitotic divisions in the meristematic tissues - the apical and the lateral meristem, result in a continuous growth of plants throughout their life.

116. Answer (3)

M-phase is the most dramatic period of cell cycle. Interphase corresponds more than 95% duration of cell cycle.

117. Answer (1)

Cells differ greatly in size, shape and activities. For example, Mycoplasmas, the smallest cells, are only  $0.3 \mu\text{m}$  in length, while bacteria could be 3 to  $5 \mu\text{m}$ . Human red blood cells are about  $7.0 \mu\text{m}$  in diameter.

118. Answer (1)

Nuclear division and cytokinesis occur twice during the process of meiosis.

119. Answer (2)

Anaphase is the best stage to study shapes of chromosomes.

Metacentric chromosome appears V-shaped during anaphase.

120. Answer (4)

Terminal ends of chromosomes are called telomeres. Centrioles take part in formation of spindle apparatus. Kinetochore are present around centromere and provide a site for attachment to spindle fibres.

121. Answer (1)

Bacterial flagellum is composed of three parts filament, hook and basal body.

122. Answer (4)

Ends of chromosomes are called telomeres. They seal the ends of chromosomes. Chromatids are held together at one point called centromere. On the sides of centromere, disc shaped structures are present known as kinetochores.

In Acrocentric chromosome, one extremely short and one very long arm is present.

123. Answer (2)

Nucleolus is a site for active ribosomal RNA synthesis.

124. Answer (3)

The isolated lysosomal vesicles have been found to be very rich in almost all types of hydrolytic enzymes optimally active at the acidic pH.

125. Answer (4)

In telophase chromosomes cluster at opposite spindle poles and their identity is lost as discrete elements.

126. Answer (2)

Pachytene stage, crossing over leads to recombination of genetic material.

127. Answer (2)

Chiasmata forms in diplotene stage. Cells at the end of prophase, when viewed under the microscope, do not show golgi complexes, ER, nucleolus and the nuclear envelope.

128. Answer (2)

The prophase I is subdivided into five stages which are arranged in following sequence.

Leptotene, Zygotene, Pachytene, Diplotene, Diakinesis.

129. Answer (3)

Cell theory was given by Schleiden and Schwann. Golgi apparatus was first observed by Camillo Golgi in 1898. Ribosomes were discovered by George Palade. Rudolf Virchow first explained that cells divided and new cells are formed from pre-existing cells (*Omnis cellula-e-cellula*).

130. Answer (4)

Meiosis increases the genetic variability in the population of organisms from one generation to the next. Variation are very important for the process of evolution.

131. Answer (2)

The complete disintegration of the nuclear envelope marks the start of second phase of mitosis (i.e., metaphase)

132. Answer (1)

Pumps are proteins that use energy to transport substances across the cell membrane.

133. Answer (4)

In oocytes of some vertebrates, diplotene can last for months or years.

134. Answer (2)

*Ulothrix*, a green alga, is a eukaryotic organism.

135. Answer (1)

Anaphase is the best stage of the cell cycle to study shape of the chromosomes.

#### SECTION - B

136. Answer (4)

There is no membrane boundary for a centriole, but it is surrounded by pericentriolar satellites, also known as massule (microtubule generator or MTG)

137. Answer (4)

Mitochondria are the sites of aerobic respiration. They produce cellular energy in the form of ATP, hence they are called 'Power houses' of the cell.

138. Answer (2)

Aleuroplasts store proteins while elaioplasts store oils and fats.

139. Answer (2)

In prophase, the DNA strands get untangled to form compact mitotic chromosome.

140. Answer (3)

A tetrad has four chromatids but they are of two homologous chromosomes.

141. Answer (3)

In plant cell middle lamella is chiefly made up of calcium pectate.

142. Answer (1)

In some organisms, karyokinesis is not followed by cytokinesis as a result of which multinucleate condition arises leading to formation of syncytium.

143. Answer (3)

The given figure is of fluid mosaic model of plasma membrane.

A - Sugar

- B - Peripheral protein  
C - Phospholipid bilayer  
D - Integral protein

144. Answer (3)

Membrane proteins can be classified as integral and peripheral proteins depending on the ease of extraction.

145. Answer (3)

Eukaryotes show intracellular compartmentalisation *i.e.*, membrane bound cell organelles.

146. Answer (2)

The interphase lasts more than 95% of the total duration of cell cycle.

When duration of cell cycle is 120 minutes then its duration of interphase should be more than 95% of 120 minutes, *i.e.*, = more than 114 minutes

147. Answer (2)

During mitosis, number of chromosomes in parent and progeny cells remain the same.

Hence, it is also called equational division.

148. Answer (2)

Since,  $n = 4$  (given)

$$\therefore 2n = 8$$

Therefore, the cell has 4 bivalents on equatorial plate during metaphase-I.

Bivalent is the complex formed by a pair of synapsed homologous chromosomes.

149. Answer (4)

- In S-phase, centriole duplicates in cytoplasm of animal cells.
- Centrioles are absent in plant cell.
- DNA replication occurs in S-phase.
- In  $G_2$  phase of cell cycle, semi-autonomous cell organelles duplicates and tubulin proteins are synthesized.

150. Answer (1)

During mitotic anaphase the centromere splits, chromatids separate and move to the opposite poles.

## ZOOLOGY

### SECTION - A

151. Answer (3)

Humans have two lungs which are covered by double layered pleura, with pleural fluid between them. The outer pleural membrane is in close contact with the thoracic lining whereas the inner pleural membrane is in contact with the lung surface.

Thoracic vertebrae are the structural constituents of the thoracic cage.

152. Answer (4)

Exchange of  $O_2$  and  $CO_2$  at the sites of exchange of gases is dependent on concentration gradient, solubility of the gases and on the thickness of diffusion membrane.

Rate of diffusion of one gas is independent from the rate of diffusion of another gas.

153. Answer (2)

In a healthy adult man, under normal physiological conditions,

TV = approx. 500 mL

IRV = 2500 mL to 3000 mL

ERV = 1000 mL to 1100 mL

RV = 1100 mL to 1200 mL

TV + ERV + IRV is less than TLC and equal to VC.

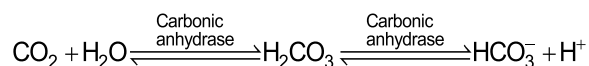
154. Answer (1)

Specialised vascularised structures called gills (branchial respiration) are used by fishes (*e.g.*, *Pterophyllum*), most of the aquatic arthropods (*e.g.*, Prawn) and molluscs (*e.g.*, *Sepia*).

Earthworms use their moist cuticle and insects have a network of tubes (tracheal tubes) for respiration.

155. Answer (4)

RBCs contain a very high concentration of the enzyme, carbonic anhydrase. This enzyme facilitates the following reaction in both the directions.



$Zn^{2+}$  is the co-factor of carbonic anhydrase.

156. Answer (3)

In the alveoli, where there is high  $pO_2$ , low  $pCO_2$ , lesser  $H^+$  concentration and lower temperature, all are the favourable factors for the formation of oxyhaemoglobin.

157. Answer (2)

Pulmonary artery carries deoxygenated blood whereas pulmonary vein carries oxygenated blood.

| Respiratory Gas | Atmospheric Air | Alveoli | Blood (Deoxygenated) | Blood (Oxygenated) | Tissues |
|-----------------|-----------------|---------|----------------------|--------------------|---------|
| O <sub>2</sub>  | 159             | 104     | 40                   | 95                 | 40      |
| CO <sub>2</sub> | 0.3             | 40      | 45                   | 40                 | 45      |

158. Answer (4)

Every 100 mL of oxygenated blood can deliver around 5 mL of O<sub>2</sub> to the tissues under normal physiological conditions in an adult human.

Therefore, 300 mL of oxygenated blood will deliver around 15 mL of O<sub>2</sub> to the tissues.

Also, every 100 mL of deoxygenated blood deliver approximately 4 mL of CO<sub>2</sub> to the alveoli.

So, 400 mL of deoxygenated blood will deliver around 16 mL of CO<sub>2</sub> to the alveoli.

159. Answer (4)

During forced expiration, contraction of internal inter-costal muscles occurs leading to the pulling of ribs in downward and inward direction, decreasing the volume of thoracic cavity.

As the volume of thoracic cavity decreases, the pressure inside thoracic cavity increases than the atmospheric pressure, leading to expiration.

160. Answer (1)

The part starting with the external nostrils up to the terminal bronchioles constitute the conducting part of human respiratory system whereas alveoli and their ducts form the exchange part of the respiratory system. The trachea, primary, secondary and tertiary bronchi and initial bronchioles are supported by incomplete cartilaginous rings.

161. Answer (2)

*Erythroblastosis foetalis* can be avoided by administering anti-Rh antibodies to the mother immediately after the delivery of the first child.

162. Answer (4)

Simple organisms like sponges (*e.g.*, *Sycon*) and coelenterates (*e.g.*, *Hydra* and *Obelia*) circulate water from their surroundings through their body cavities to facilitate the cells to exchange essential substances. Complex organisms (*e.g.*, *Pheretima*) use special fluids within their bodies for such exchange.

163. Answer (2)

About 70 per cent filling of ventricles takes place during joint diastole. Both the atria undergo a simultaneous contraction during atrial systole. This increases the flow of blood into the ventricles by about 30%.

164. Answer (3)

Plasma is a straw coloured, viscous fluid constituting nearly 55 per cent of the blood and formed elements constitute about 45 per cent of the blood.

165. Answer (1)

Clot/coagulum is formed by a network of threads called fibrins which are derived from inactive fibrinogens.

166. Answer (4)

In fishes, the heart pumps out deoxygenated blood which is oxygenated by the gills and supplied to the body parts from where deoxygenated blood is returned to the heart (single circulation).

167. Answer (1)

The nodal musculature has the ability to generate action potentials without any external stimuli, *i.e.*, it is autoexcitable.

However, the number of action potentials that could be generated in a mixture vary at different parts of the nodal system. The SAN can generate the maximum number of action potentials *i.e.*, 70-75 min<sup>-1</sup> and is responsible for initiating and maintaining the rhythmic contractile activity of the heart.

168. Answer (3)

As the blood passes through the capillaries in tissues, some water along with many water soluble substances move out into the spaces between the cells of tissues leaving the larger proteins and most of the formed elements in the blood vessels. This released out fluid is called lymph.

169. Answer (3)

Basophils are the cells responsible for secretion of histamine and serotonin.

Monocytes and lymphocytes are the agranulocytes.

170. Answer (4)

Fishes have a 2-chambered heart with an atrium and a ventricle. In fishes, blood get oxygenated by gills and heart pumps out deoxygenated blood.

In humans, oxygenated and deoxygenated blood received by the left and right atrium respectively, passes into the ventricles of the same side which pump it out without any mixing up.

171. Answer (4)

Silicosis is common occupational respiratory disorder. In asthma, there is difficulty in breathing causing wheezing due to inflammation of bronchi and bronchioles.

172. Answer (2)

A sigmoid curve is obtained when percentage saturation of haemoglobin with  $O_2$  is plotted against the  $pO_2$ .

This sigmoid curve is known as oxygen dissociation curve.

173. Answer (2)

The epithelial tissue present in the alveoli of human lungs is simple squamous epithelium.

If the cuboidal or columnar cells bear cilia on their free surface they are called ciliated epithelium. Their function is to move particles or mucus in a specific direction over the epithelium. They are mainly present in the inner surface of hollow organs like bronchioles.

174. Answer (2)

Neural signals through the sympathetic nerves can increase the rate of heart beat, the strength of ventricular contraction and thereby the cardiac output.

Also, adrenal medullary hormones like epinephrine and norepinephrine can increase the cardiac output.

175. Answer (3)

A bicuspid/Mitral valve guards the opening between the left atrium and the left ventricle. Mitral valve will evert out into atrium during ventricular systole in this condition.

176. Answer (2)

Blood vessels in between left ventricle and right atrium constitute systemic circulation through which blood flows.

177. Answer (1)

A bundle of nodal fibres, AV bundle continues from the AVN which passes through the atrio-ventricular septa to emerge on the top of the interventricular septum and immediately divides into a right and left bundle, which transmit action potential to the entire ventricular musculature through purkinje fibers.

178. Answer (2)

Upon holding the breath, the amount of  $CO_2$  in the body increases, which leads to a change in the pH of the blood.

This changing pH gets recognised by the chemoreceptors present in blood vessels activates respiratory rhythm centre in brain, signals the muscles associated with respiratory system and leads to an urge to breath.

179. Answer (4)

Tunica intima is the inner layer of blood vessels made up of squamous endothelium. The smallest vessels called capillaries have walls which are one-celled thick.

180. Answer (1)

A specialised nodal tissue embedded in the lower corner of the right atrium, close to atrio-ventricular septum, delays the spreading of impulses to heart apex for about 0.1 seconds. This delay allows the atria to empty completely.

181. Answer (3)

Heart failure means the state of heart when it is not pumping blood effectively enough to meet the needs of the body.

It is not same as cardiac arrest (when heart stops beating) or a heart attack (when the heart muscle is suddenly damaged by an inadequate blood supply).

182. Answer (3)

An injury or trauma stimulates the platelets in blood to release certain factors which activate the mechanism of coagulation. Platelets, also known as thrombocytes are 1.5 lakh - 3.5 lakh per  $mm^3$  of blood.

183. Answer (4)

Factors for clotting of blood are present in the plasma in an inactive form. Plasma without the clotting factors is called serum.

184. Answer (1)

In a standard ECG, the P-wave represents the electrical excitation (or depolarisation of the atria), the QRS complex represents the depolarisation of the ventricles. The T-wave represents the repolarisation of ventricles and end of T-wave marks the end of the ventricular systole.

185. Answer (4)

In an open type of blood circulation, the blood is pumped out of the heart and the cells and tissues are directly bathed in it.

Hemichordates, arthropods, non-cephalopod molluscs have an open type of blood circulation whereas most of annelids and chordates have closed type of blood circulation.

#### SECTION - B

186. Answer (3)

A special neural centre in the medulla oblongata can moderate the cardiac function through autonomic nervous system (ANS).

187. Answer (1)

Cardiac output =  $SV \times HR$

$$\begin{aligned}\text{Stroke volume} &= \text{EDV} - \text{ESV} \\ &= 150 \text{ mL} - 100 \text{ mL} \\ &= 50 \text{ mL}\end{aligned}$$

$$\begin{aligned}\text{HR} &= \frac{\text{CO}}{\text{SV}} = \frac{5000 \text{ mL/minute}}{50 \text{ mL}} \\ &= 100 \text{ beats per minute}\end{aligned}$$

188. Answer (3)

In well-trained athletes, stroke volume can also increase significantly allowing them to achieve cardiac output during strenuous exercise upto six-seven times greater than their resting values.

189. Answer (4)

Each artery and vein consists of three layers: an inner lining of squamous endothelium, the tunica intima, a middle layer of smooth muscle and elastic fibres, the tunica media, and an external layer of fibrous connective tissue with collagen fibres, the tunica externa.

190. Answer (2)

Rh antibodies from the mother (Rh-ve) can leak into the blood of the foetus (Rh +ve) and destroy the foetal RBCs.

191. Answer (1)

- A – Squamous epithelium of alveolar wall
- B – Basement substance
- C – Red blood cell
- D – Endothelium of blood capillary

192. Answer (2)

At high altitude, atmospheric  $pO_2$  level is less and hence, more RBCs are needed to transport the required amount of  $O_2$  to the tissues to survive.

193. Answer (4)

Intrapleural fluid is present between the two pleural layers of lungs. This fluid helps in maintaining the negative pressure in the pleural cavity as during inhalation the lungs expands and does not collapse because of presence of this fluid.

194. Answer (4)

Residual volume is the volume of air remaining in the lungs even after a forcible expiration. This averages 1100 mL to 1200 mL.

195. Answer (3)

Blood is the medium of transport for  $O_2$  and  $CO_2$ . About 97% of  $O_2$  is transported by RBCs in the blood. The remaining 3% of  $O_2$  is carried in a dissolved state through the plasma. Nearly 20-25% of  $CO_2$  is transported by RBCs whereas 7% of it is carried in a dissolved state through plasma.

196. Answer (3)

A chemosensitive area, situated adjacent to the respiratory rhythm centre is highly sensitive to  $CO_2$  and hydrogen ions.

Increase in  $CO_2$  and hydrogen ions can activate this chemosensitive area. The respiratory rhythm centre is located in the medulla oblongata of the brain.

197. Answer (4)

We know, Cardiac output = Heart rate  $\times$  Stroke volume

Given, 7000 mL = Heart rate  $\times$  70 mL

$$\Rightarrow \text{Heart rate} = \frac{7000}{70} = 100$$

$$\text{Now, duration of cardiac cycle} = \frac{60}{\text{Heart rate}} = \frac{60}{100}$$

$$= 0.6 \text{ seconds}$$

198. Answer (3)

| Blood Group | Antigens on RBCs | Antibodies in Plasma | Donor's Group |
|-------------|------------------|----------------------|---------------|
| A           | A                | anti-B               | A, O          |
| B           | B                | anti-A               | B, O          |
| AB          | A, B             | nil                  | AB, A, B, O   |
| O           | nil              | anti-A, B            | O             |

199. Answer (2)

During atrial systole, both bicuspid/Mitral and tricuspid valves are open but during ventricular contraction, they get close.

Also, during ventricular diastole AV valves open. Semilunar valves open when ventricles undergo contraction and are located between right ventricle and pulmonary artery and between left ventricle and aorta respectively.

200. Answer (4)

A unique vascular connection exists between the digestive tract and liver called hepatic portal system. The hepatic portal vein carries blood from intestine to the liver before it is delivered to the systemic circulation. It ensures that the food ingested is processed by the liver first before entering the systemic circulation. This way, ingested toxins are detoxified by hepatocytes.